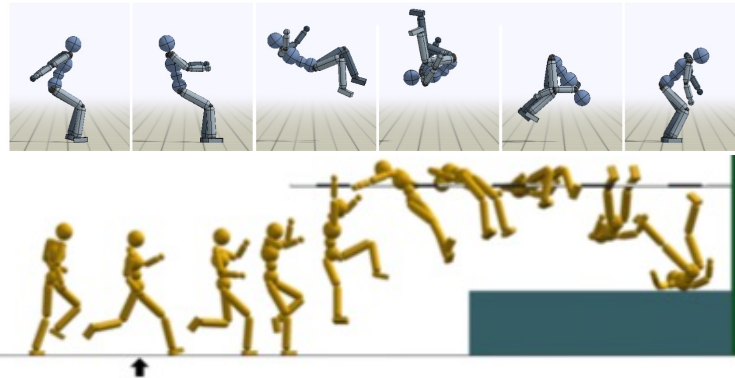




REINFORCEMENT LEARNING FOR ROBOTICS AND REALISTIC ANIMATED CHARACTERS

Michiel van de Panne

Dept of Computer Science
University of British Columbia



ROBOTICS / GRAPHICS / BIOMECHANICS: A SHARED HISTORY

- homogeneous transformation matrices
- efficient physics-based simulation
- skilled movement
- trajectory optimization
- leveraging ML/RL methods

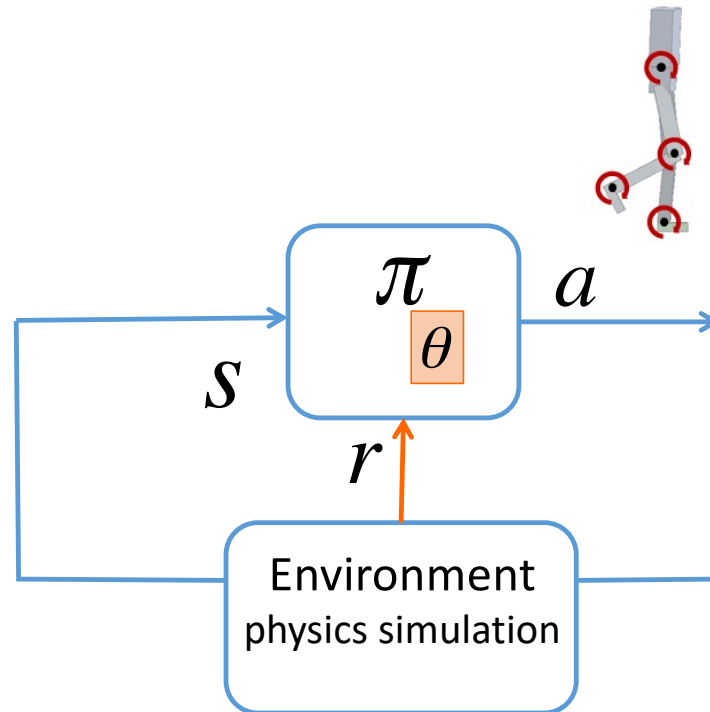
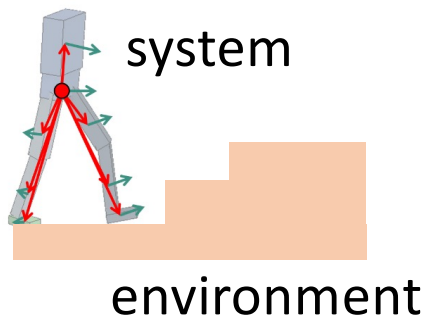


RL FOR ROBOTICS – THE HYPE CURVE



[Wikipedia, “Hype Cycle”]

REINFORCEMENT LEARNING FOR MOTION CONTROL

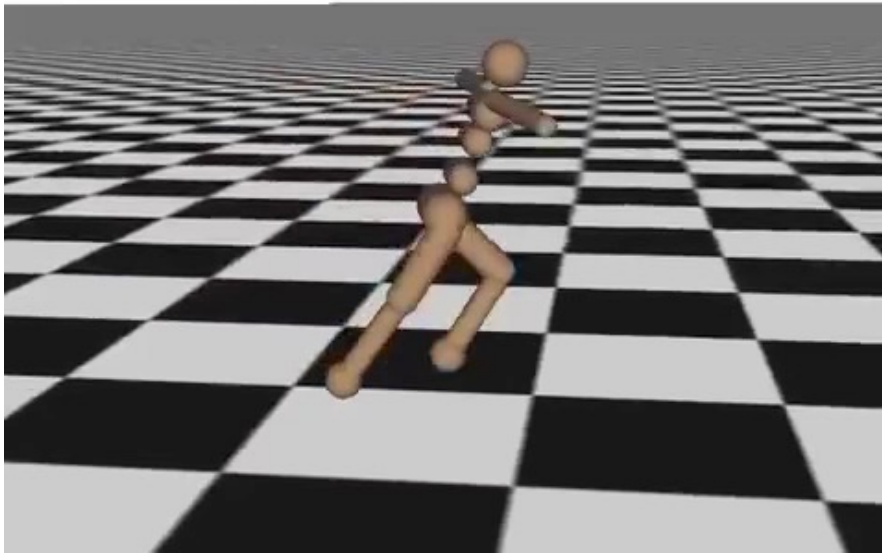


In principle:

- specify rewards
- trial-and-error learning

$$\max_{\theta} \mathbb{E} \left[\sum_{t=0}^H R(s_t) | \pi_{\theta} \right]$$

CHALLENGES: MOTION QUALITY

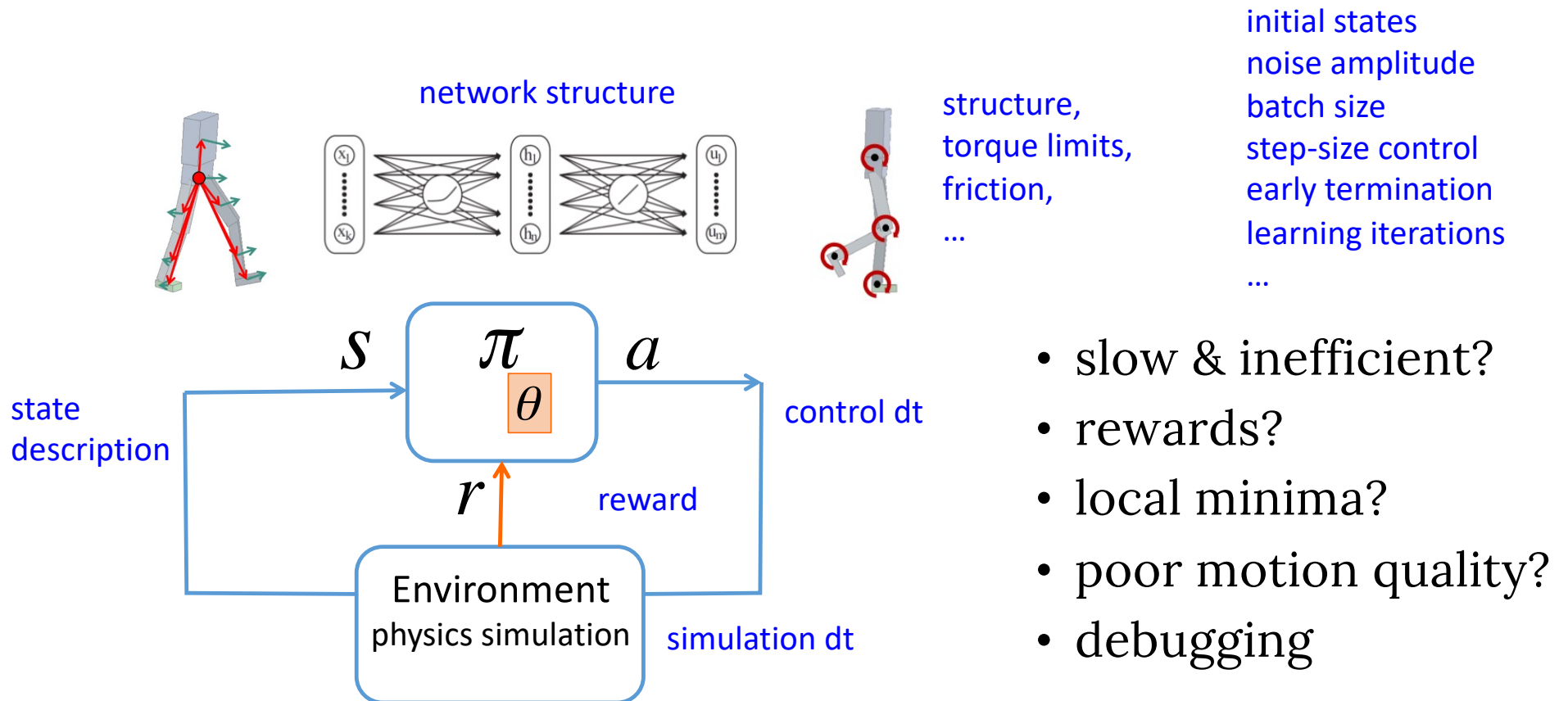


[Schulman et al. 2016]



[Heess et al. 2017]

CHALLENGES: GETTING IT TO WORK FOR YOUR PROBLEM



DEEPLOCO: DYNAMIC LOCOMOTION SKILLS USING HIERARCHICAL DEEP REINFORCEMENT LEARNING

Xue Bin Peng
Glen Berseth
KangKang Yin
Michiel van de Panne



[SIGGRAPH 2017]

DEEPMIMIC: EXAMPLE-GUIDED DEEP REINFORCEMENT LEARNING OF PHYSICS-BASED CHARACTER SKILLS

Xue Bin Peng
Pieter Abbeel
Sergey Levine
Michiel van de Panne

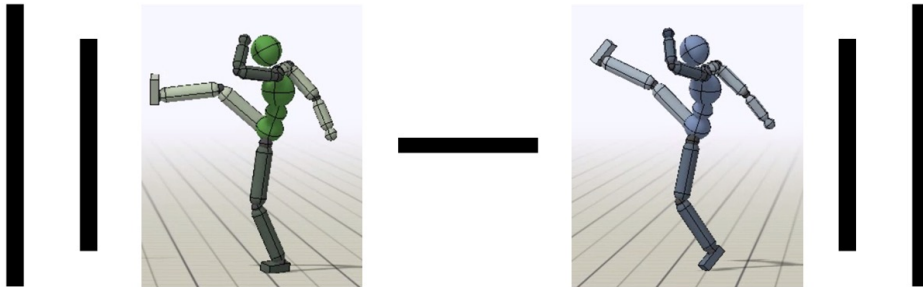


[SIGGRAPH 2018]

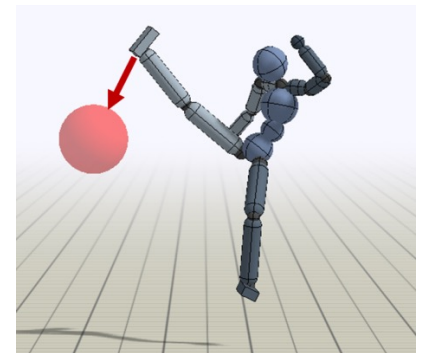
MOTION IMITATION

$$r_t = \omega^I \underline{r_t^I} + \omega^G \underline{r_t^G}$$

Imitation Objective



Task Objective



METHOD

- Policy Gradient method

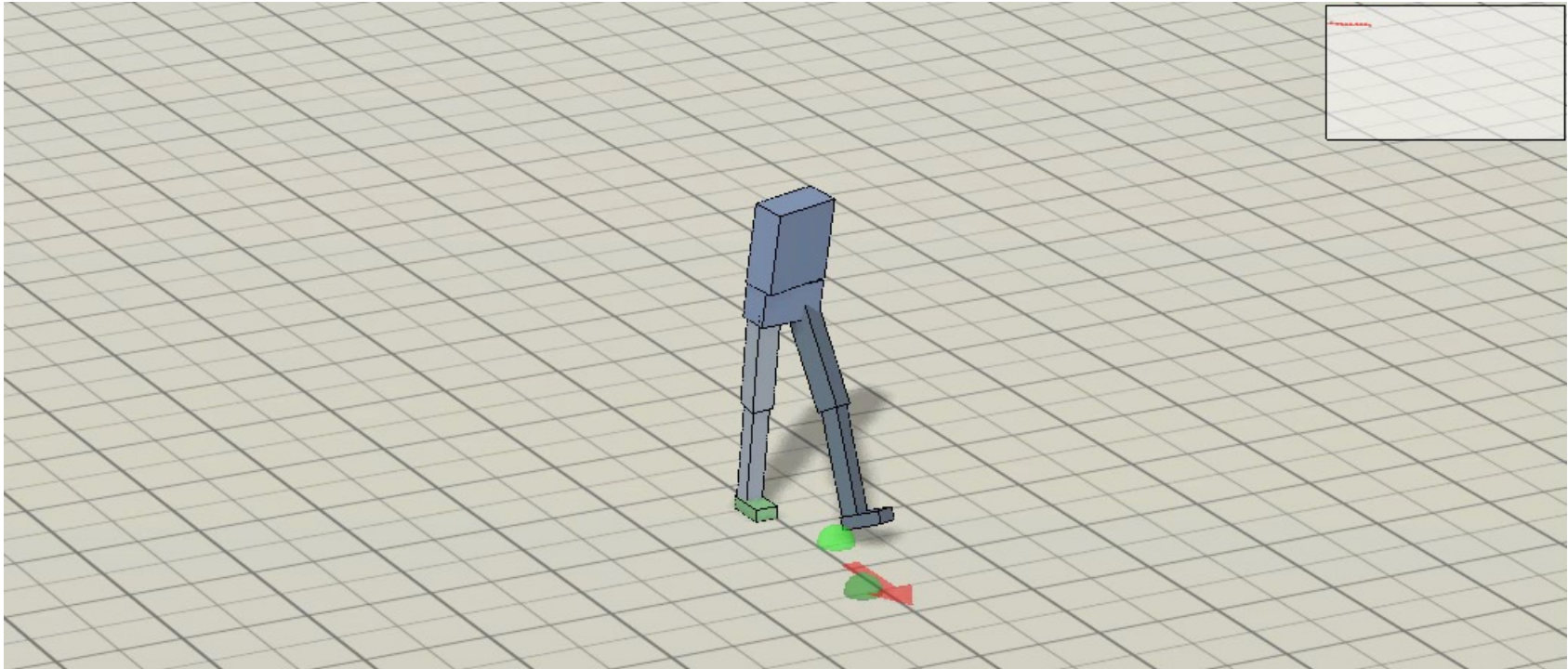
$$\nabla_{\theta} J(\theta) = \mathbb{E}_{s_t \sim d_{\theta}(s_t), a_t \sim \pi_{\theta}(a_t | s_t)} [\nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \mathcal{A}_t]$$
$$\mathcal{A}_t = R_t - V(s_t) \quad R_t = \sum_{l=0}^{T-t} \gamma^l r_{t+l}$$

generalized advantage estimator GAE(λ)

- Proximal Policy Optimization (PPO) step-size control

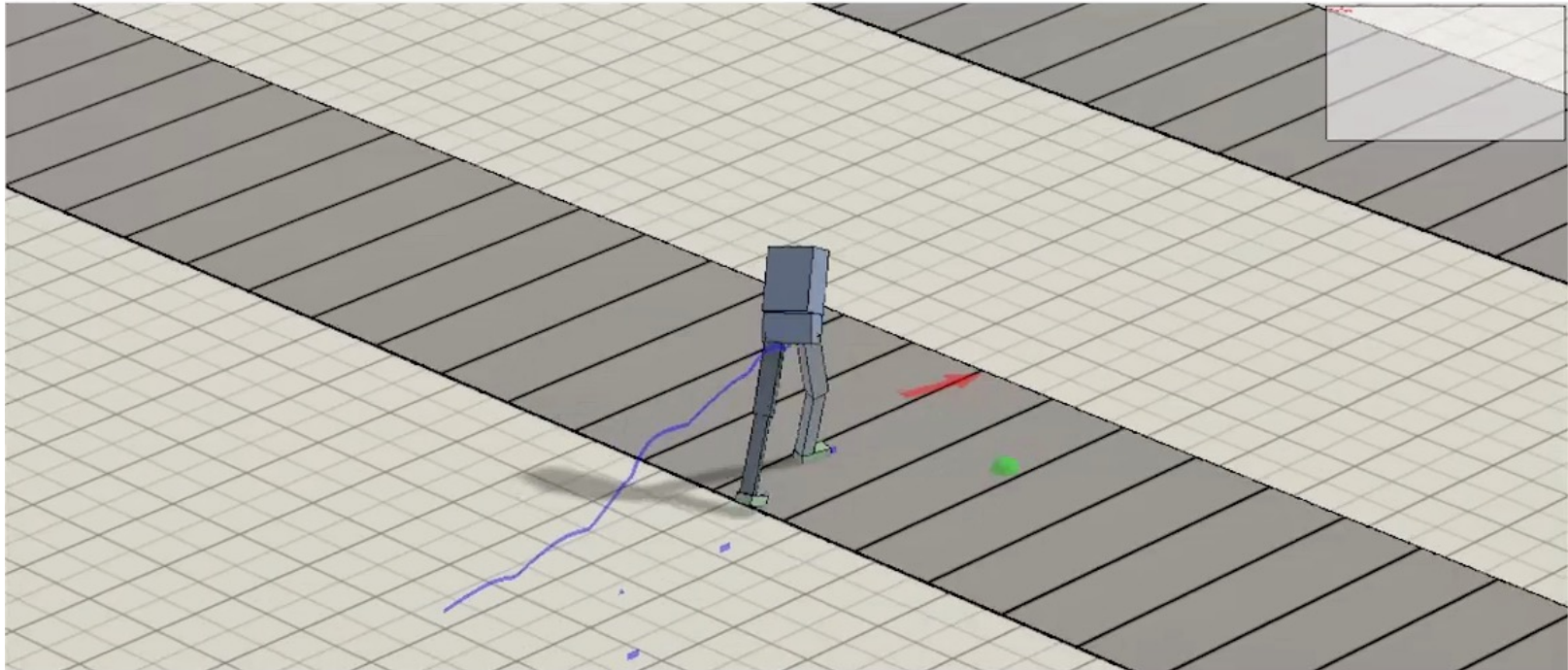
WALKING

[DeepLoco: SIGGRAPH 2017]

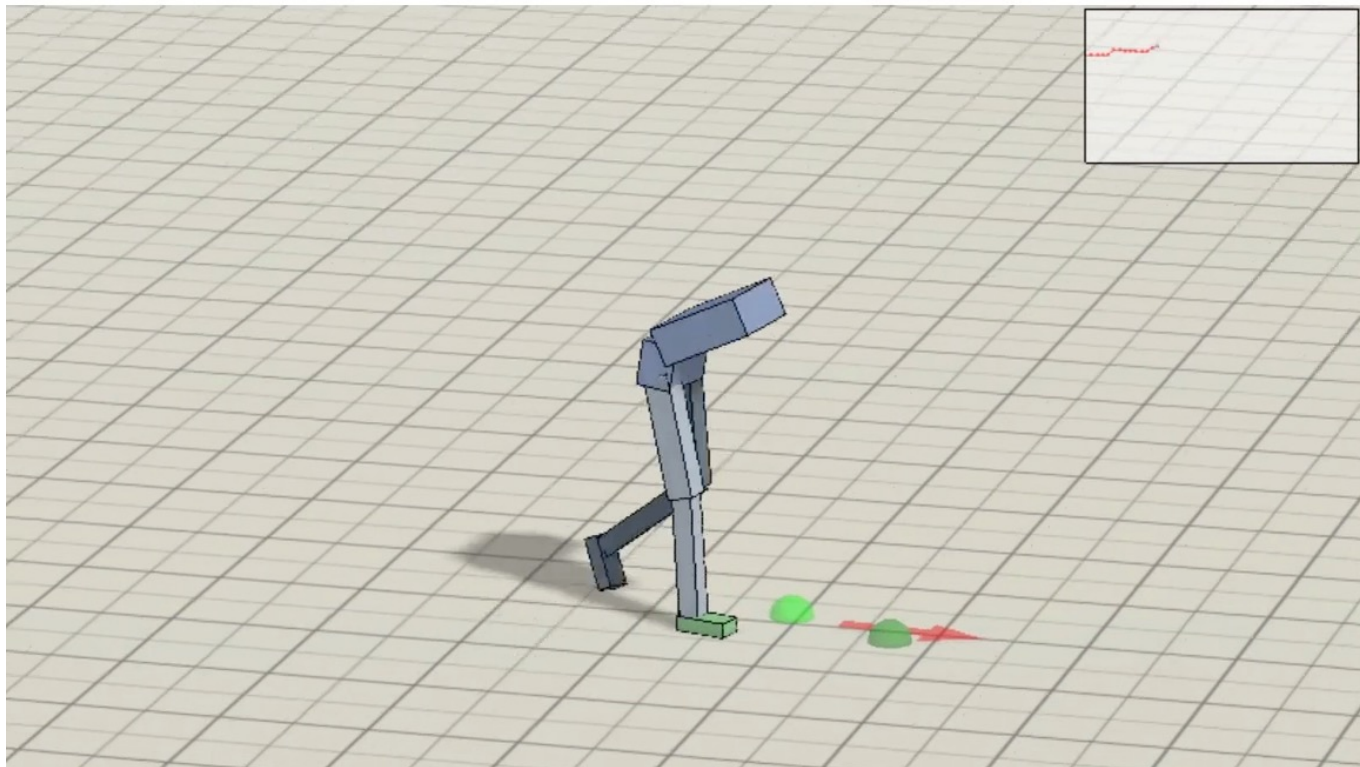


[DeepLoco: SIGGRAPH 2017]

Walking on Conveyor Belts



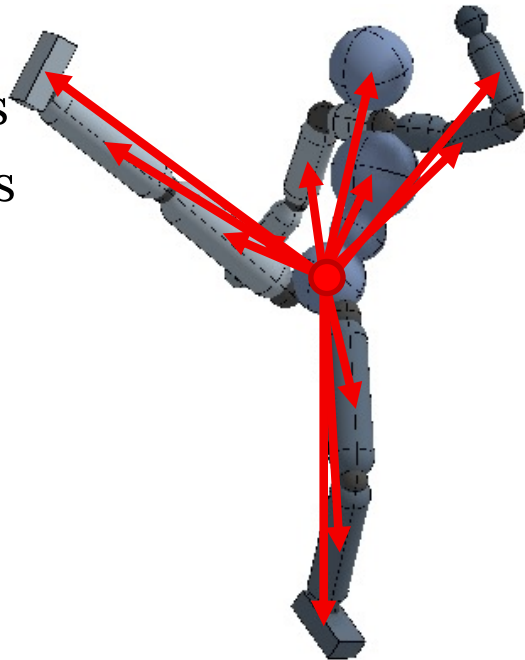
Forward Lean



MORE COMPLEX CHARACTERS & MOTIONS

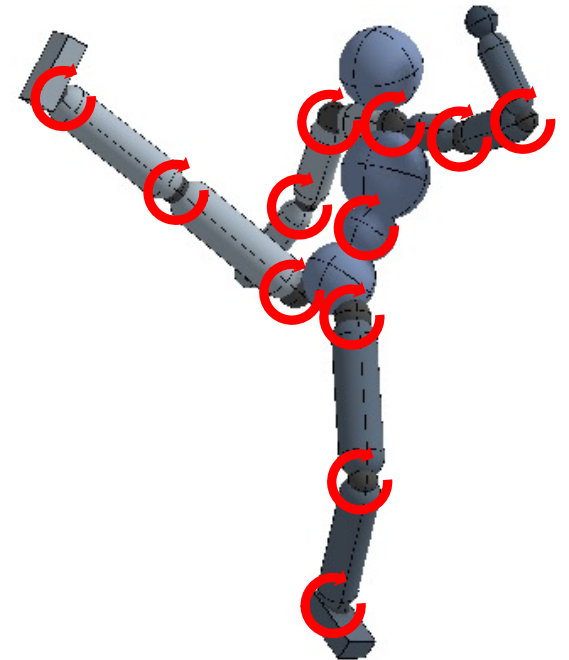
State: 197 D

- link positions
- link velocities

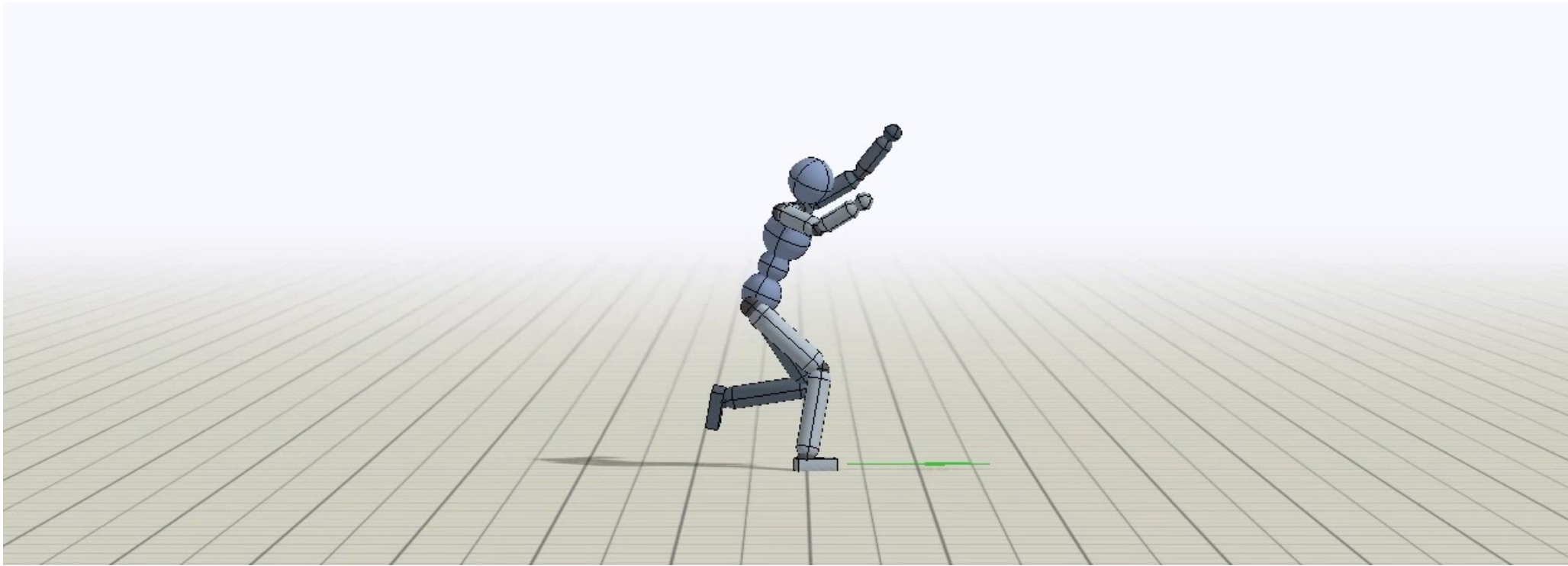


Action: 36 D

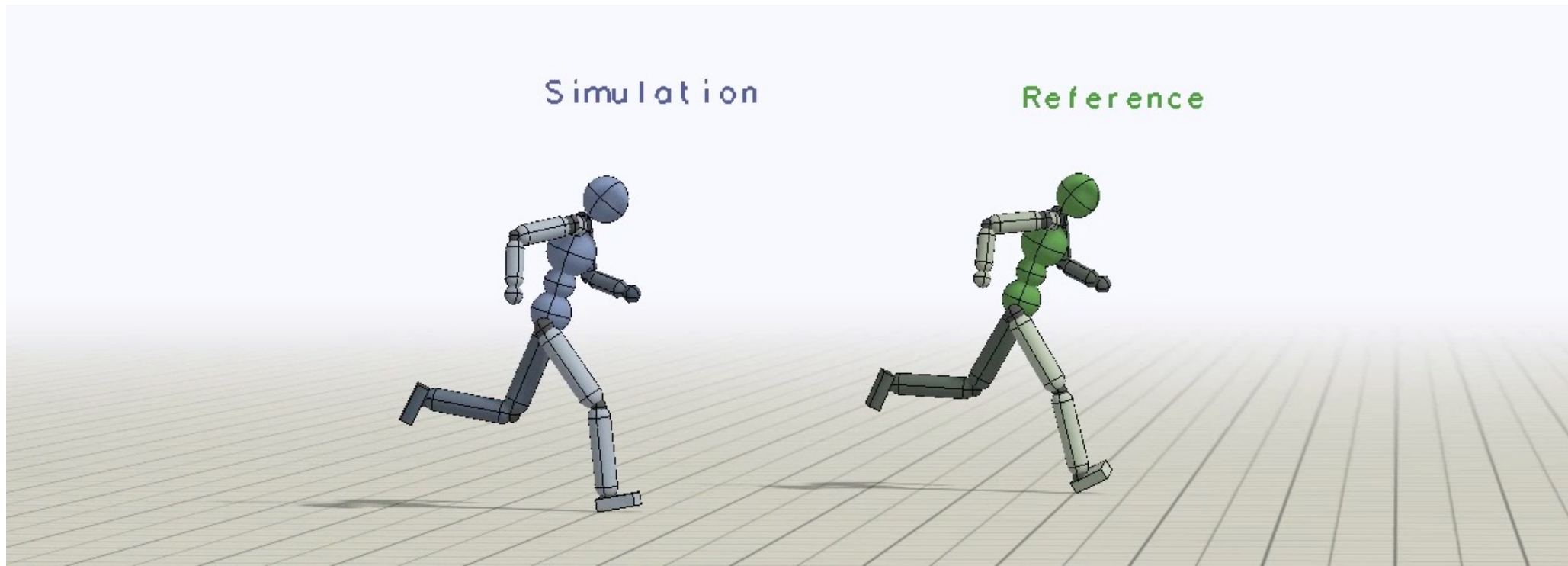
- PD targets



NO REFERENCE MOTION



HUMANOID: RUN



HUMANOID: BACKFLIP

Simulation



Reference

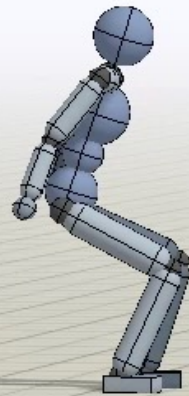


METHOD

- Proximal Policy Optimization (PPO)
- Early Termination (ET)
- Reference State Initialization (RSI)

WITH EARLY TERMINATION

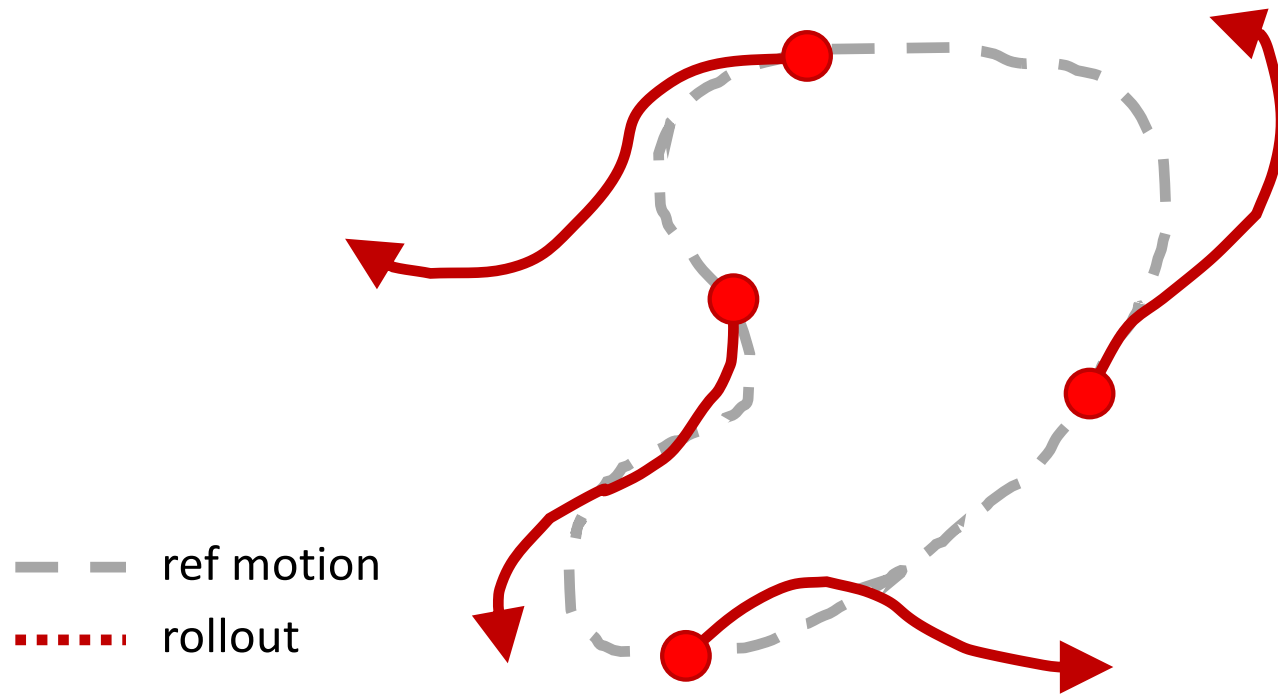
Simulation



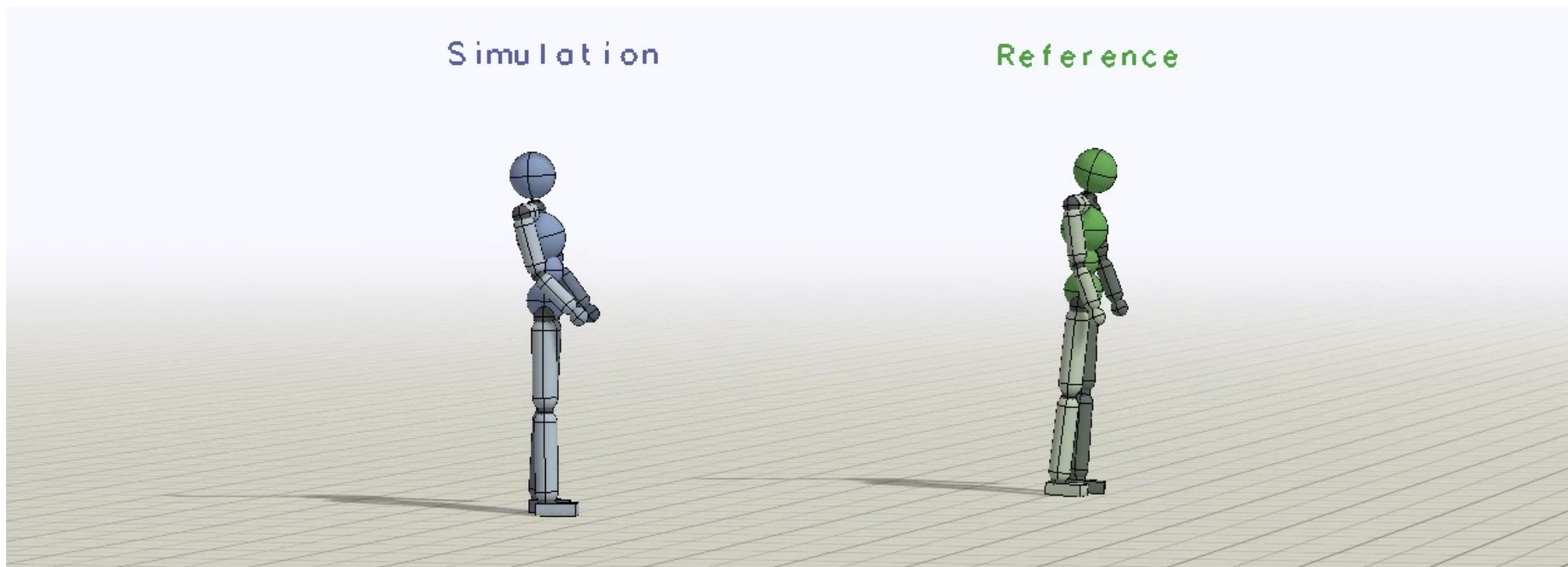
Reference



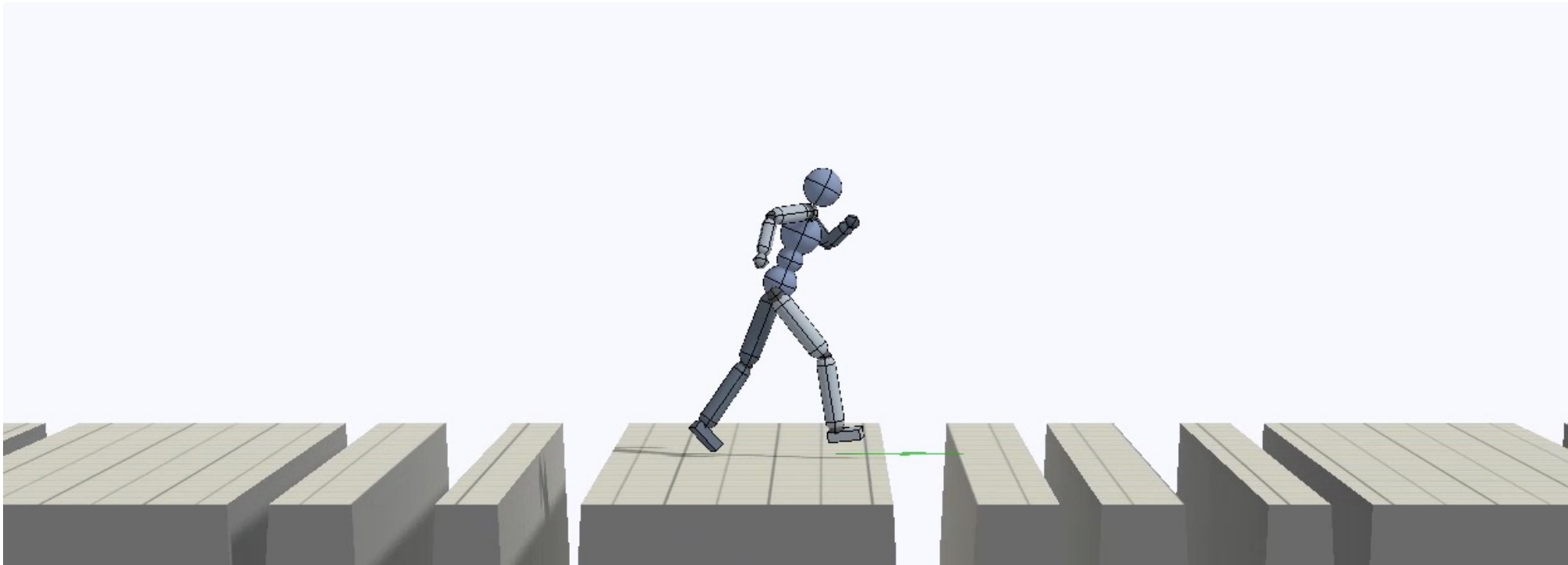
REFERENCE STATE INITIALIZATION (RSI)



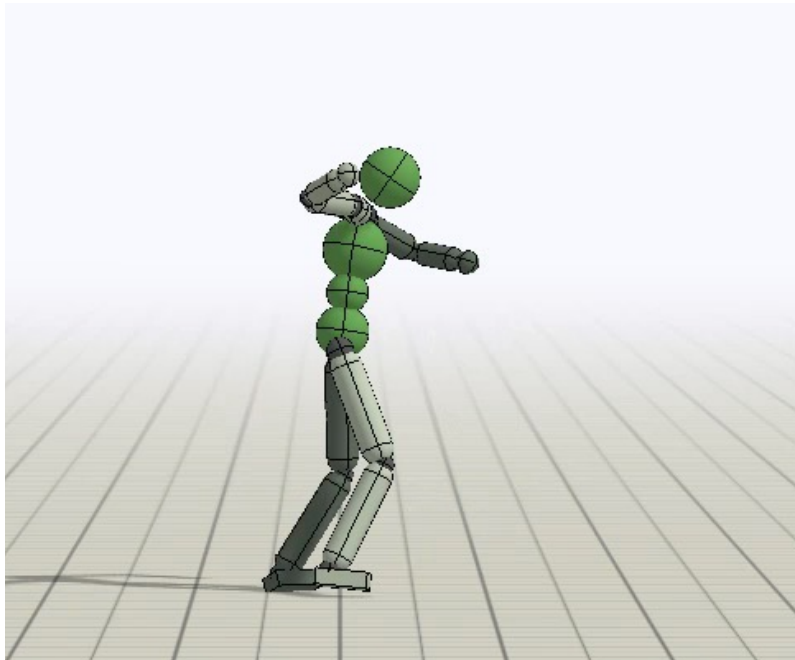
WITH REFERENCE STATE INITIALIZATION



HUMANOID: RUN – DENSE GAPS

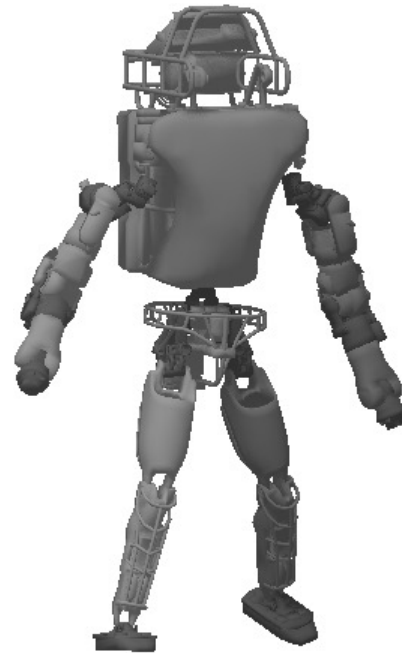


CHARACTER RETARGETING



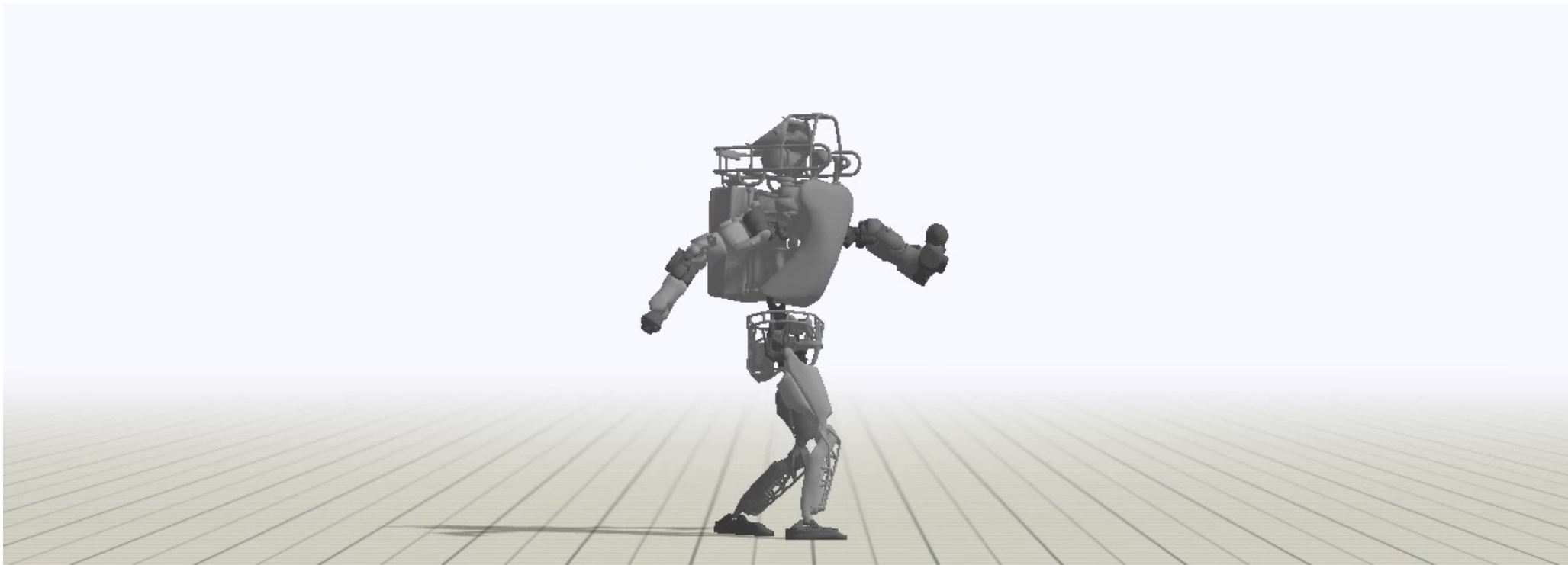
Reference Motion

+

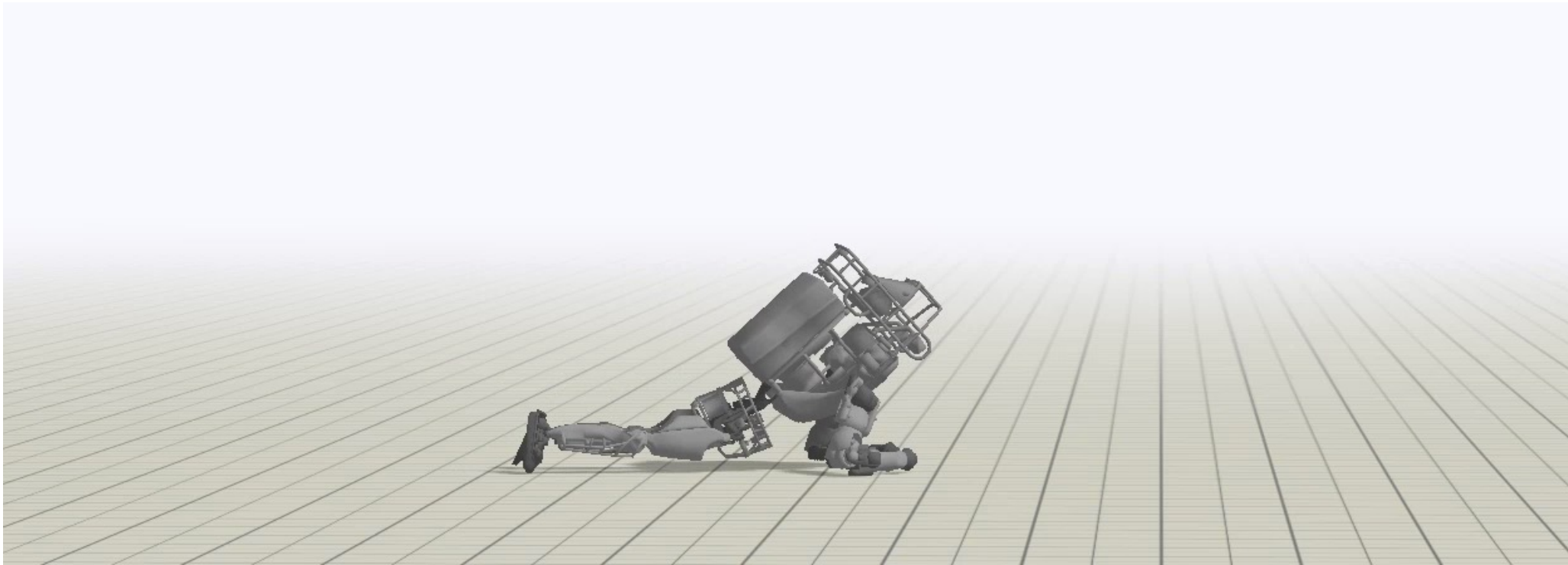


Atlas

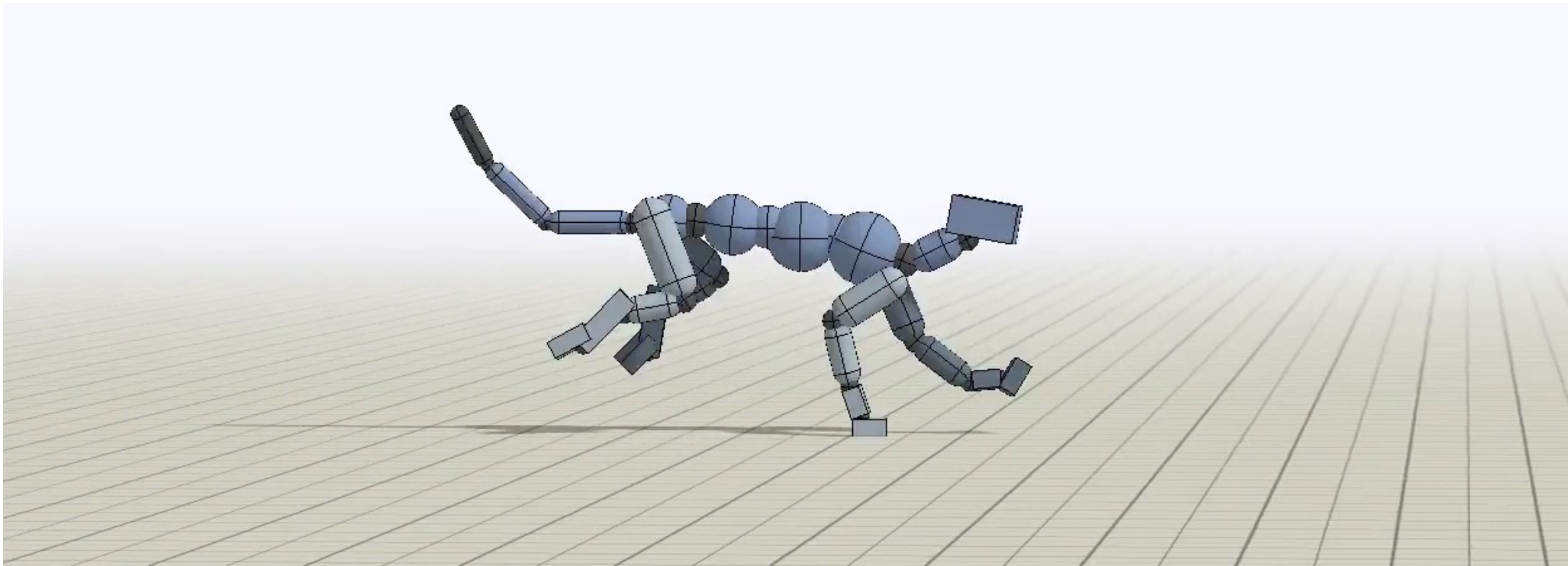
ATLAS: SPINKICK



ATLAS: GETUP- FACEDOWN



SIMULATED LION





Ziva Dynamics



ZIVA



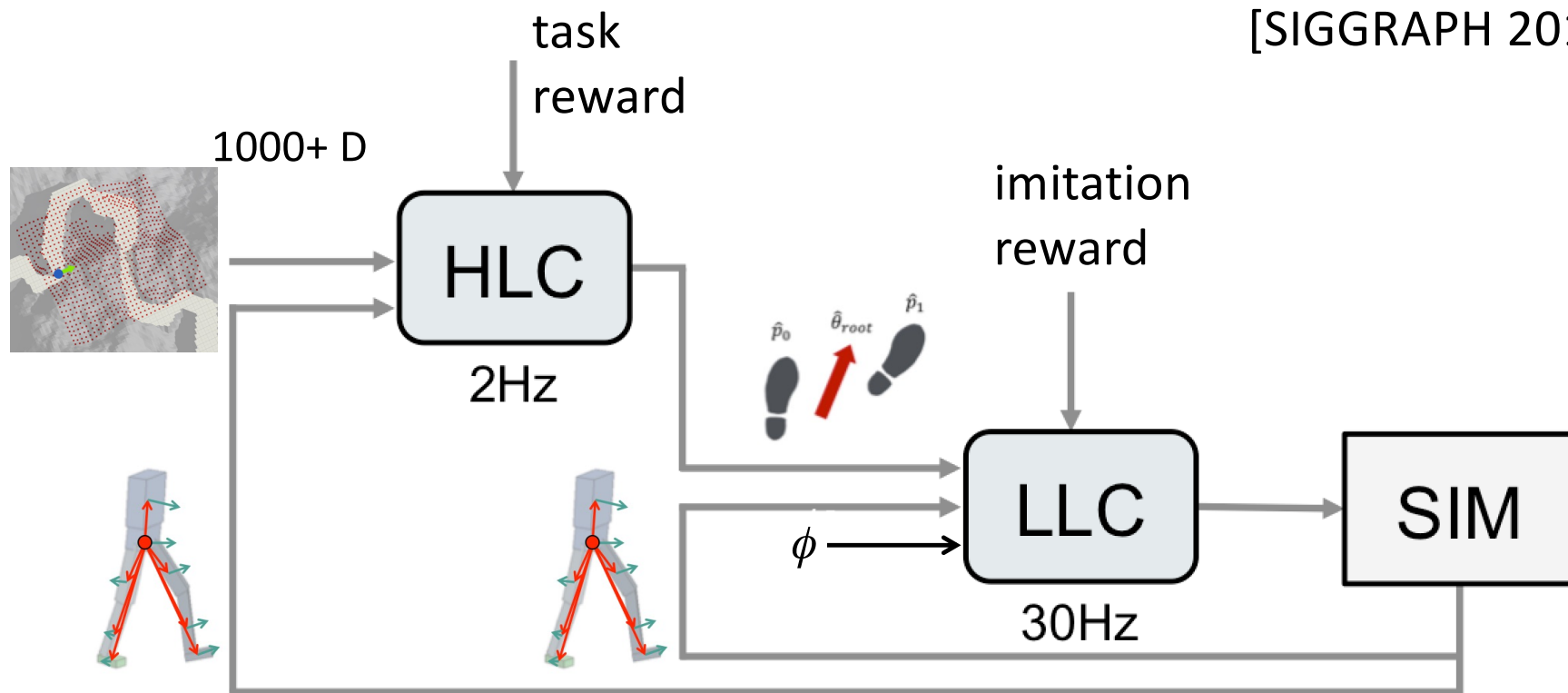
ZIVA

LESSONS LEARNED

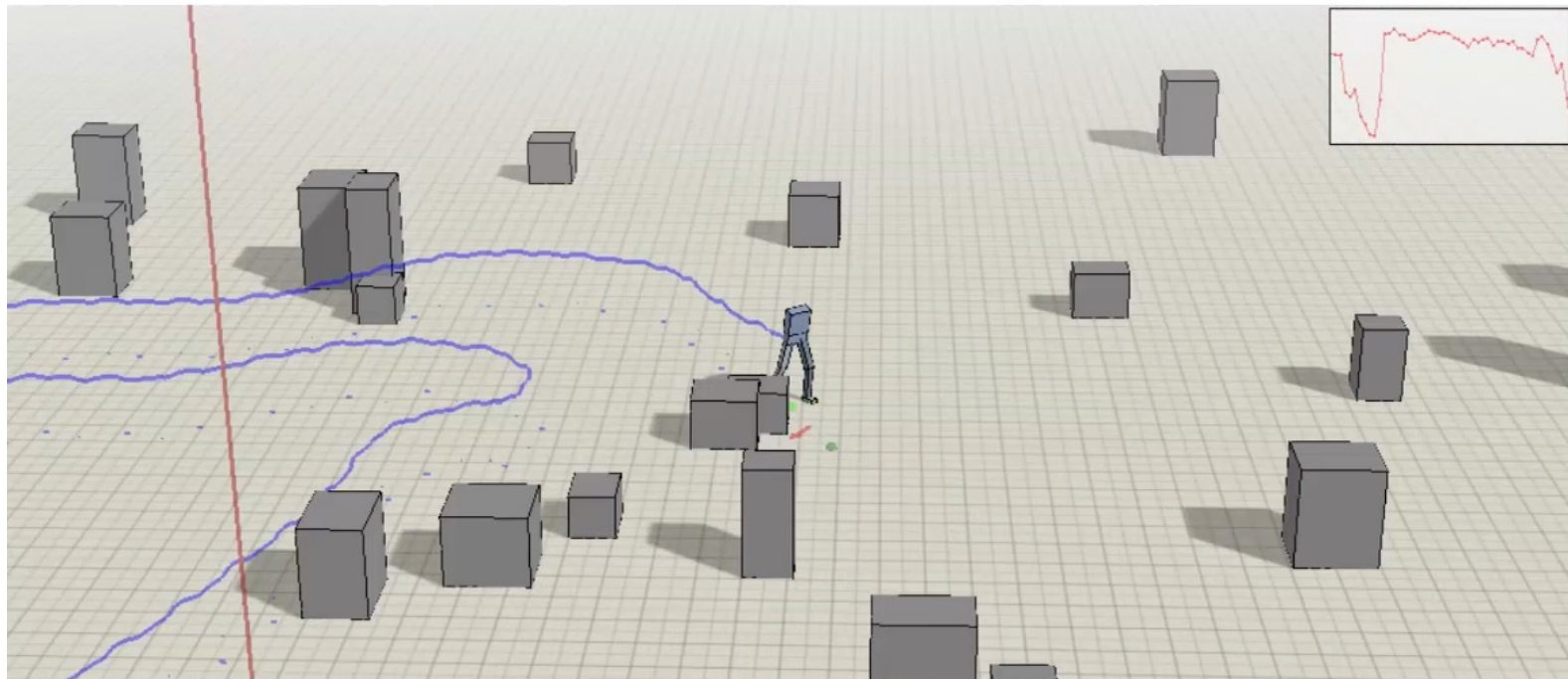
- natural and robust imitation of (nearly any) motion
- some degree of:
 - task-driven variations
 - perception-driven control
 - integration of multiple skills
- algorithmic novelty vs details, data, compute

DEEPLOCO: HIERARCHICAL RL

[SIGGRAPH 2017]

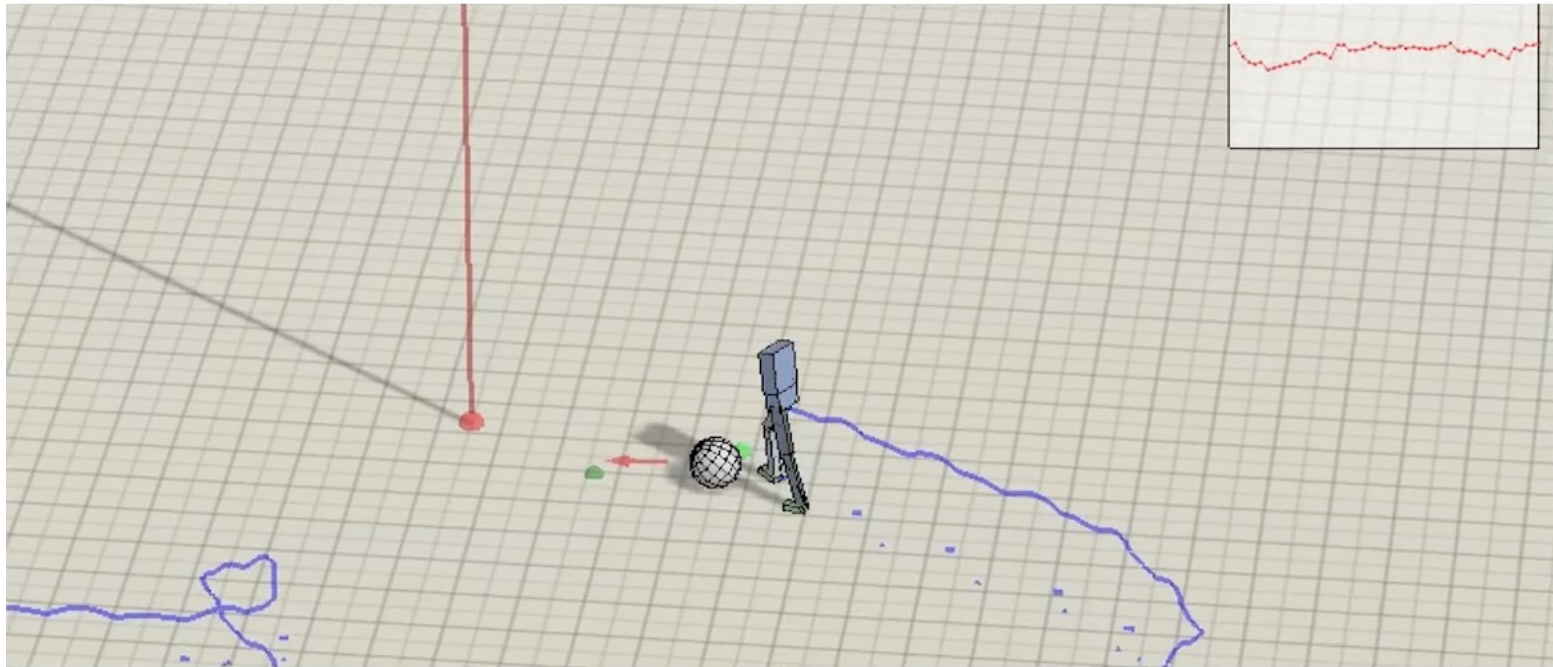


Dynamic Obstacles



HLC

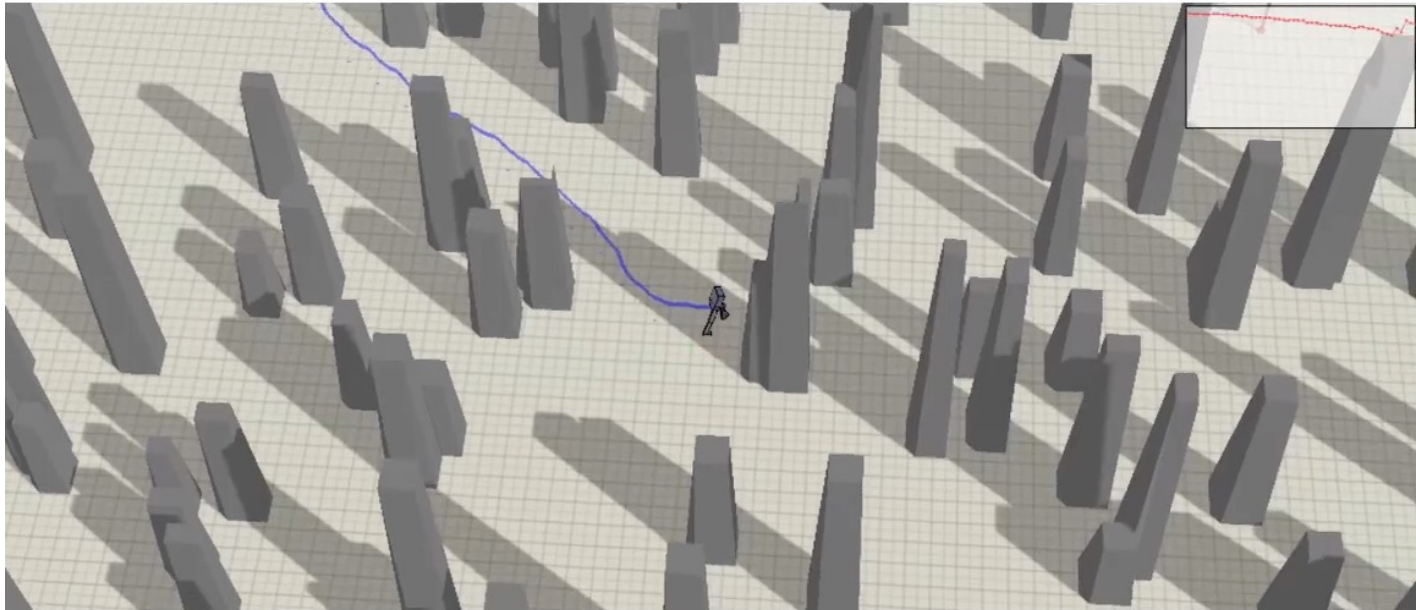
Soccer Dribbling



HLC

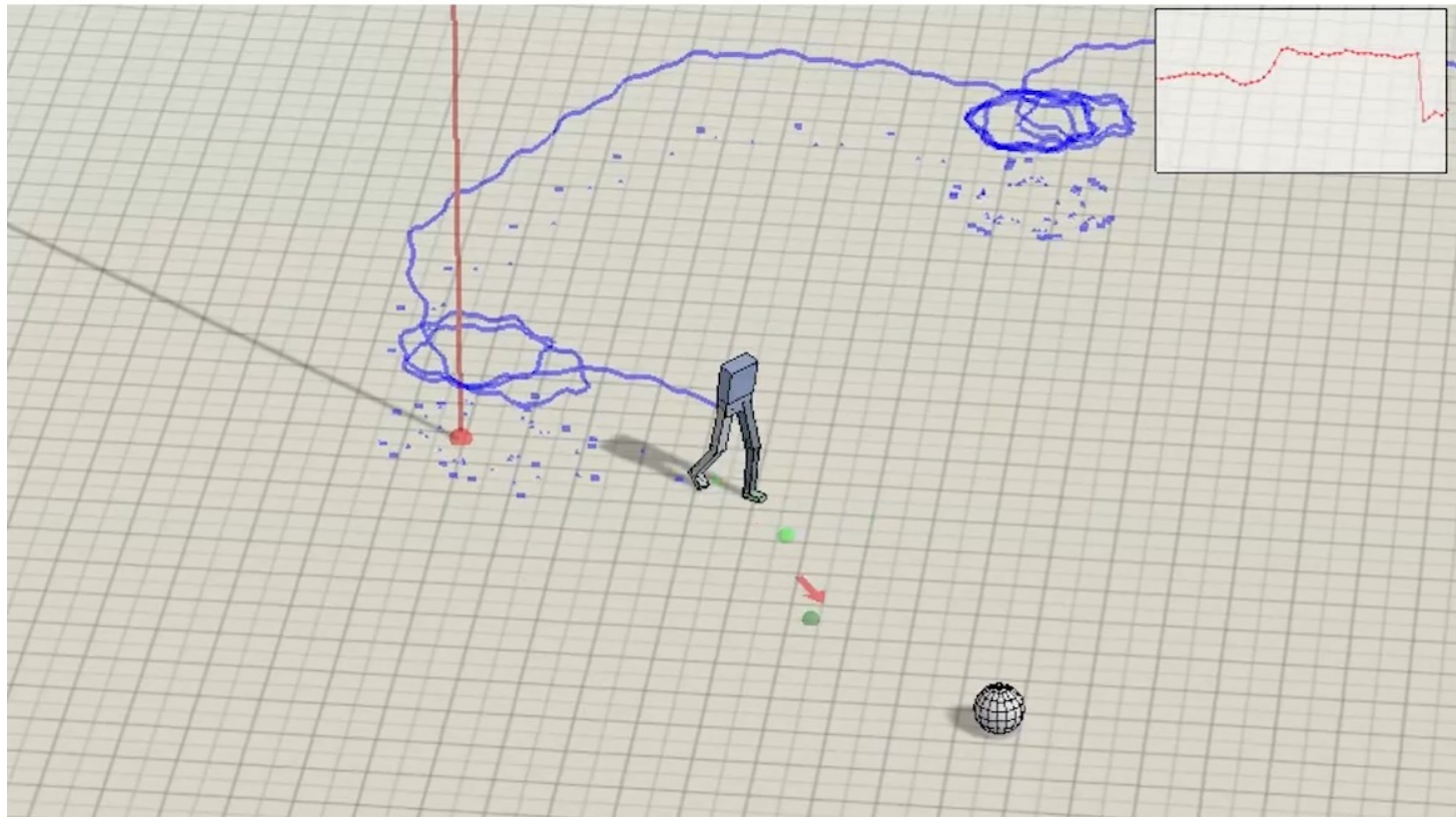
Pillars

(4x speedup)



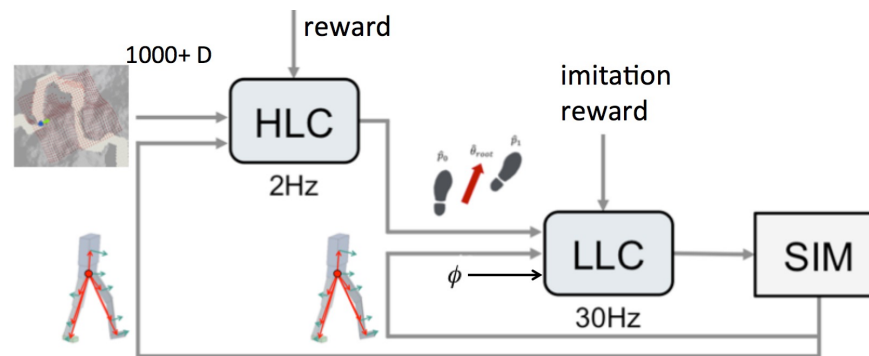
SOCCER DRIBBLING

Soccer Dribbling

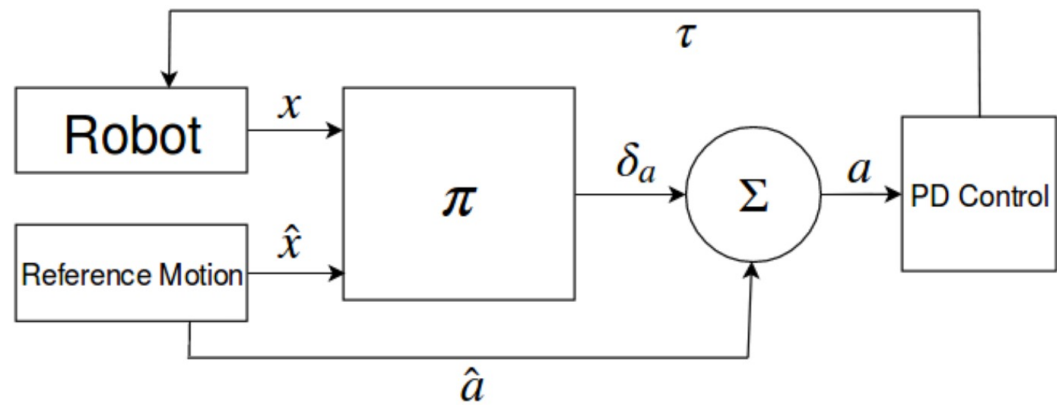
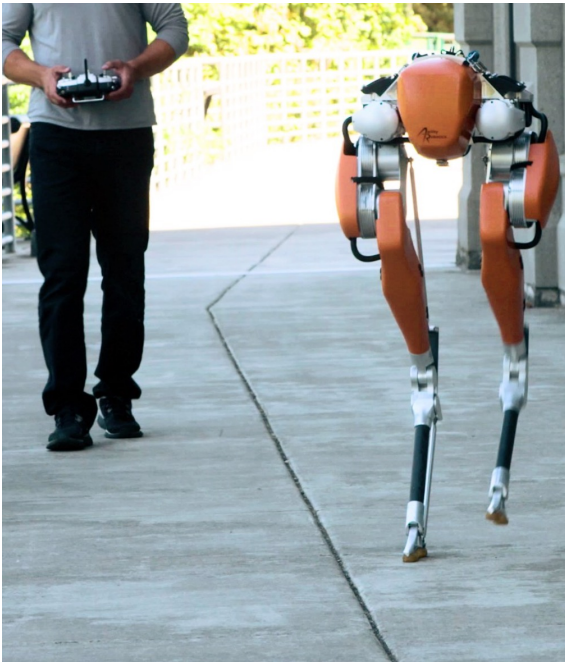


LESSONS LEARNED

- effective hierarchy
- imitation objectives
- identical algorithm for HLC, LLC learning
- robustness: self-perturbation



DO THESE METHODS EXTEND TO ROBOTS?



Cassie (Agility Robotics)
Collaboration with Oregon State University

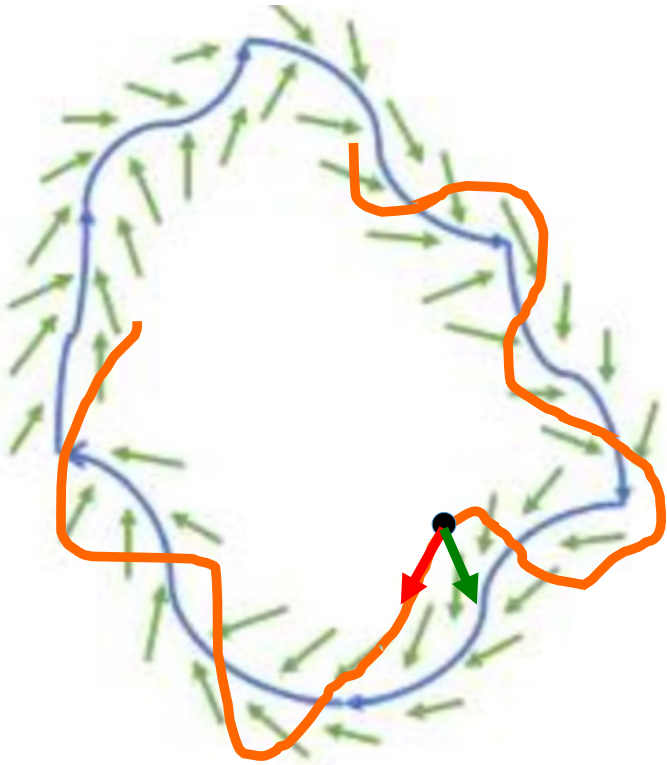
Learning Locomotion Skills for Cassie: Iterative Design and Sim-to-Real

Zhaoming Xie, Patrick Clary, Jeremy Dao, Pedro Morais, Jonathan Hurst, Michiel van de Panne
Conference on Robot Learning (CORL 2019)

MOTION DESIGN TOOLS

- motion imitation (RL)
- policy cloning (SL)
- design iteration (RL + SL)
 - design reward
 - constrained optimization
 - observe result

POLICY CLONING: GOOD SAMPLES TO LEARN FROM



(state, action)

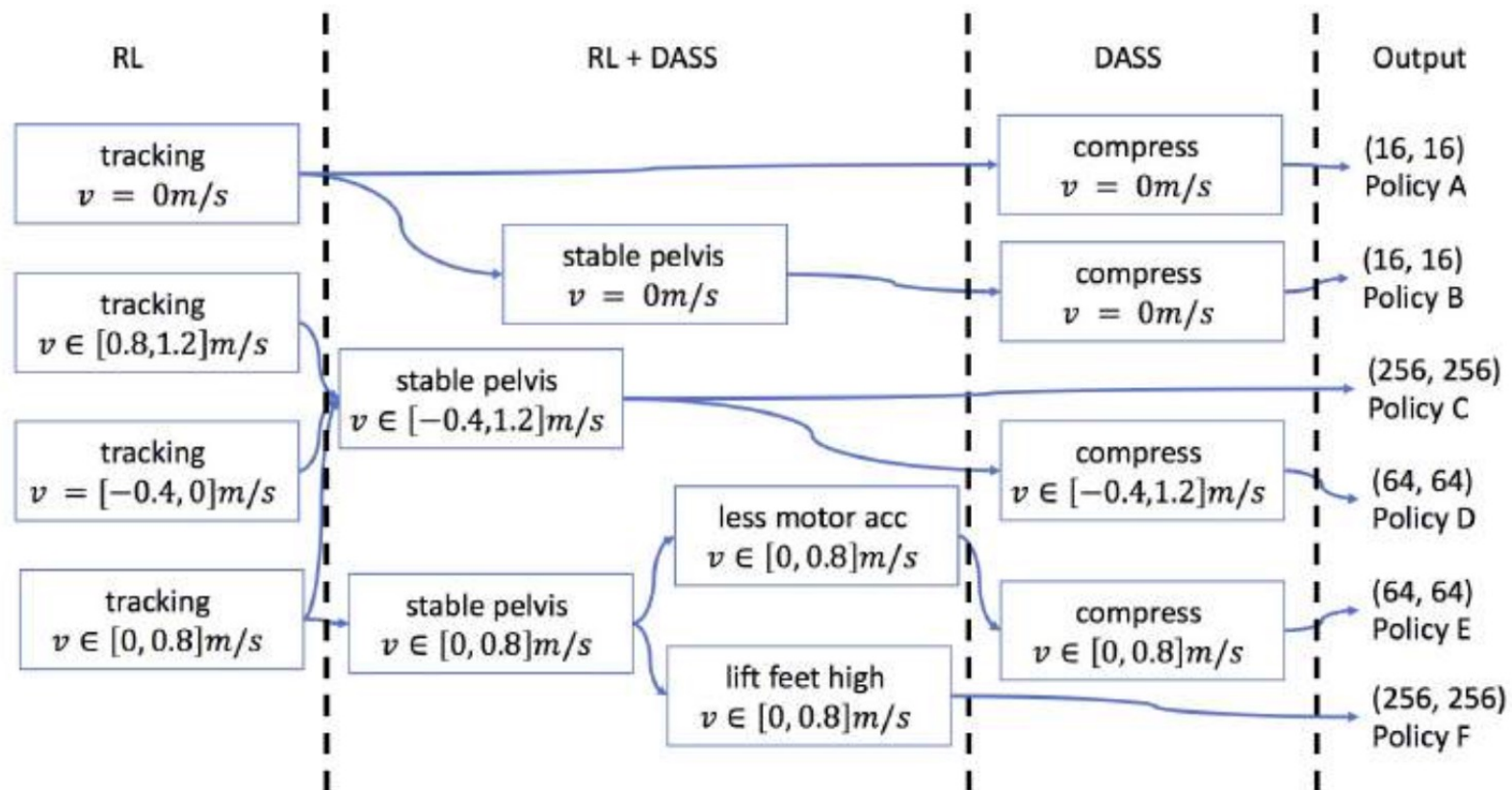
state: from stochastic policy

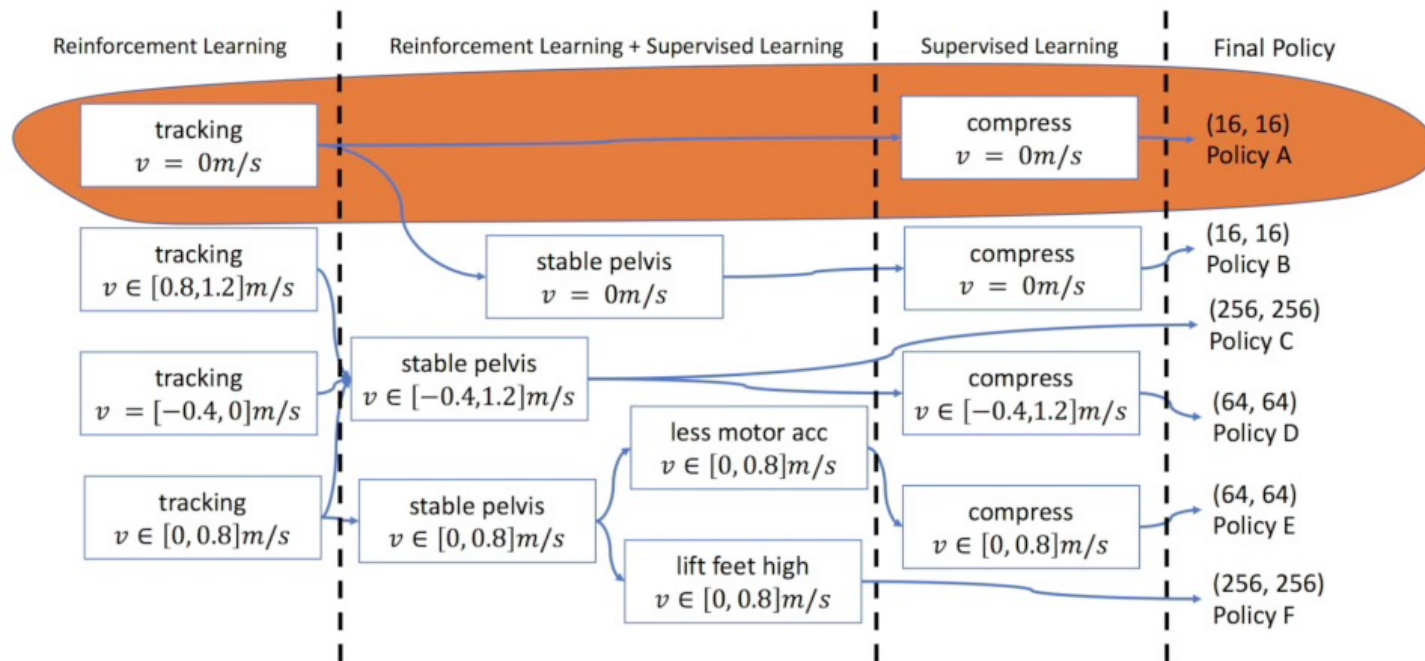
action: from deterministic policy

use for:

- merging skills
- mapping to new architecture
- constraining optimization to stay nearby

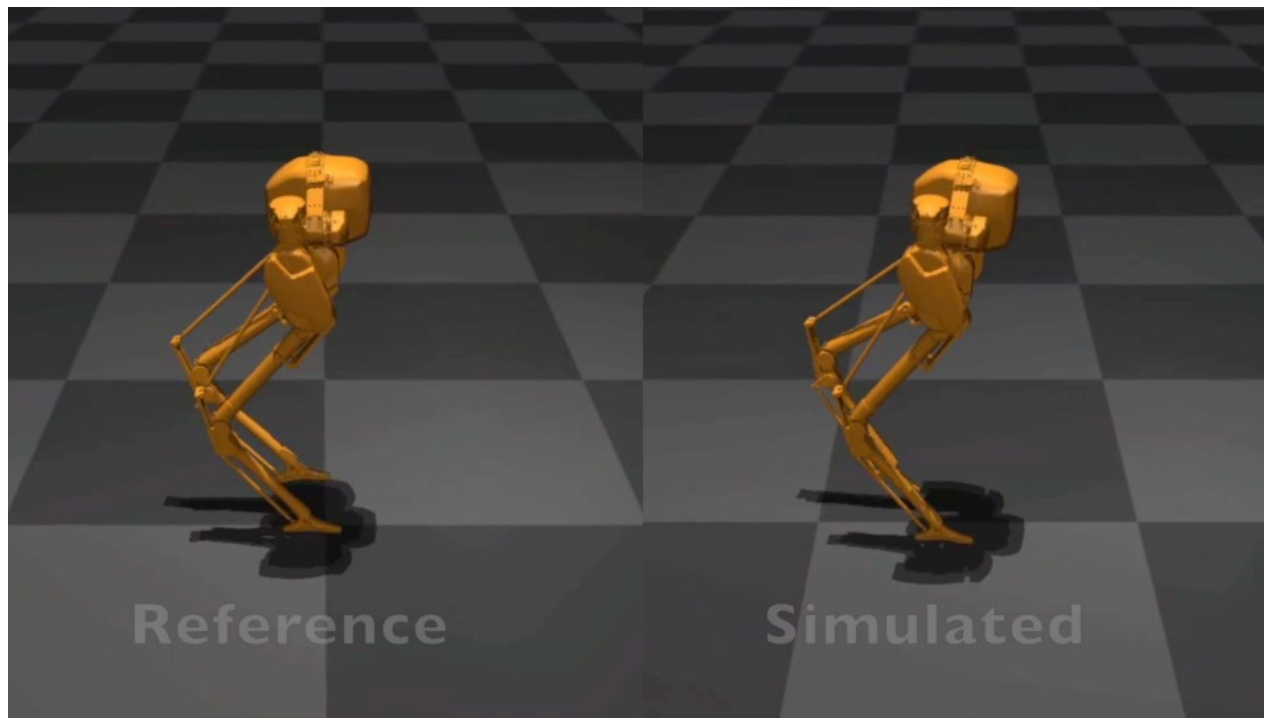
ITERATIVE DESIGN PROCESS





**Optimized policy with a naive
"forward progress" reward**

FIRST ATTEMPTS



SIM-TO-REAL V0

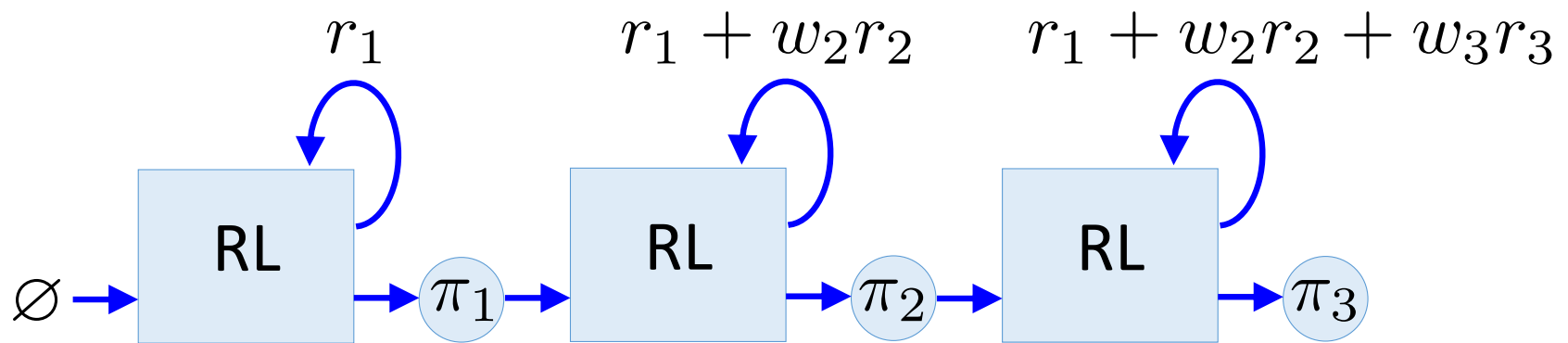


FIXES: NOTA BENE

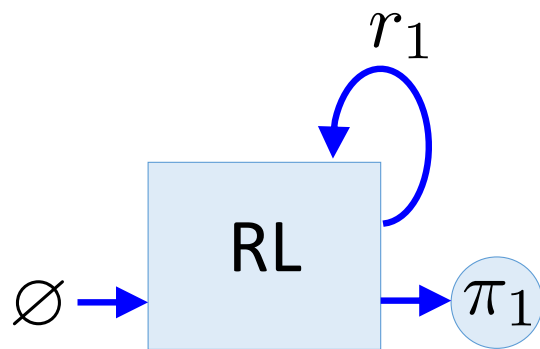
- no dynamics randomization (!)
- add symmetric tuples
- model the reflected inertia of motors
- model the delay
- train using state estimator (instead of ground truth)
- output target-delta instead of target
- policy conditioned on: state + target speed + **phase**
 - **needed for stepping in place (or condition on history)**

ITERATIVE DESIGN

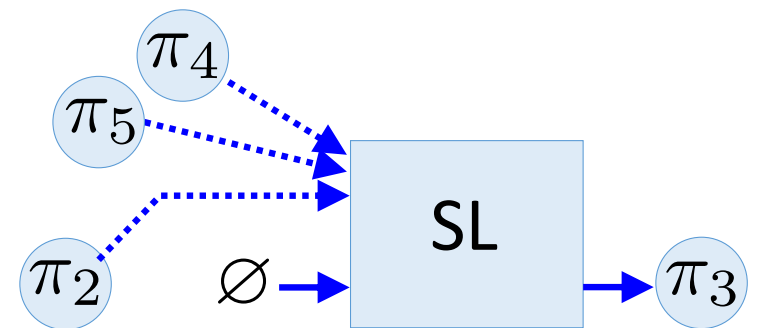
DEFAULT: REWARD TWEAKING



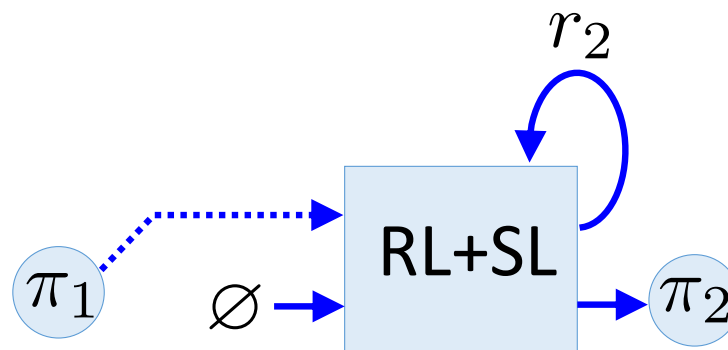
ITERATIVE DESIGN: BUILDING BLOCKS



standard RL

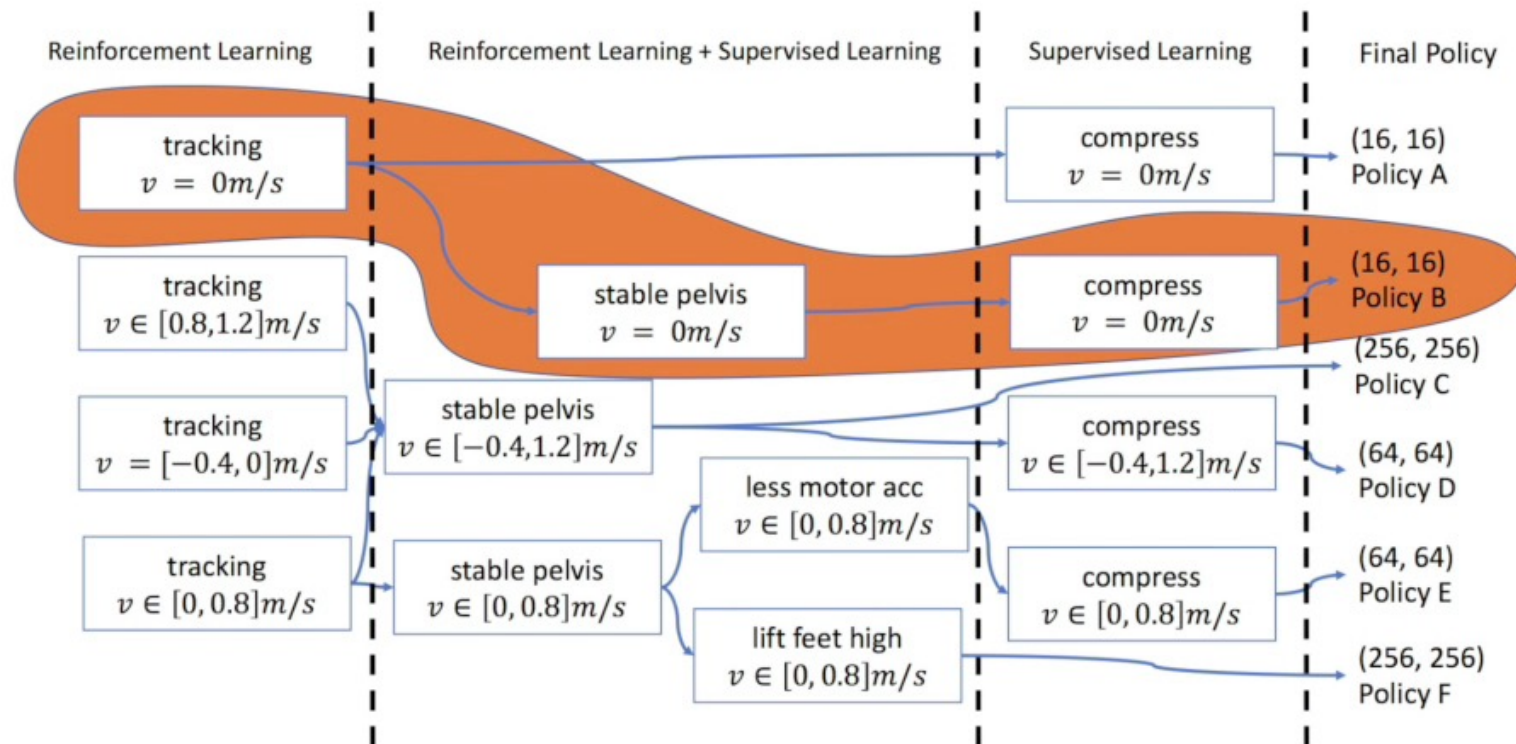


policy cloning

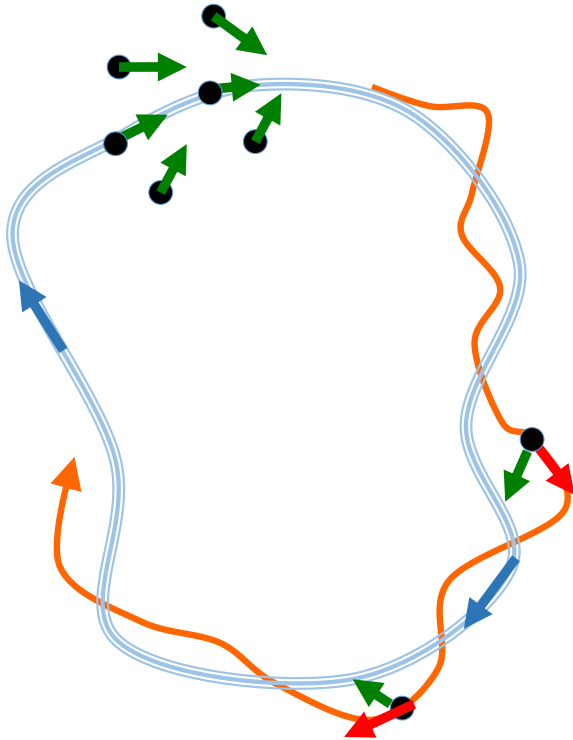


constrained RL

LEARNING FROM SMALL DATA: 600 SAMPLES, 16X16 NET



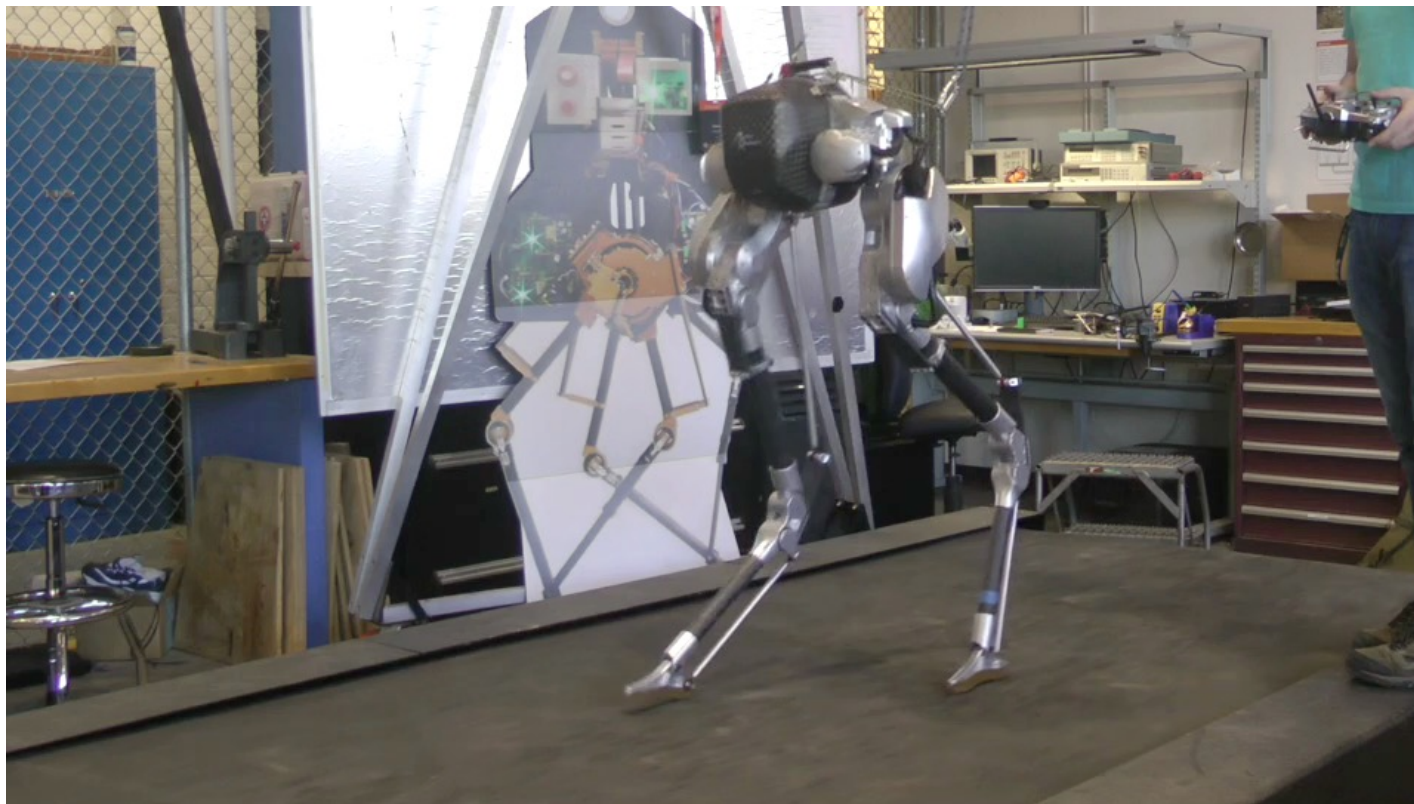
COLLECTING GOOD SAMPLES



$$\Pi = \{(s_1, a_1), (s_2, a_2), \dots\}$$

record deterministic actions for the
states visited by the stochastic policy

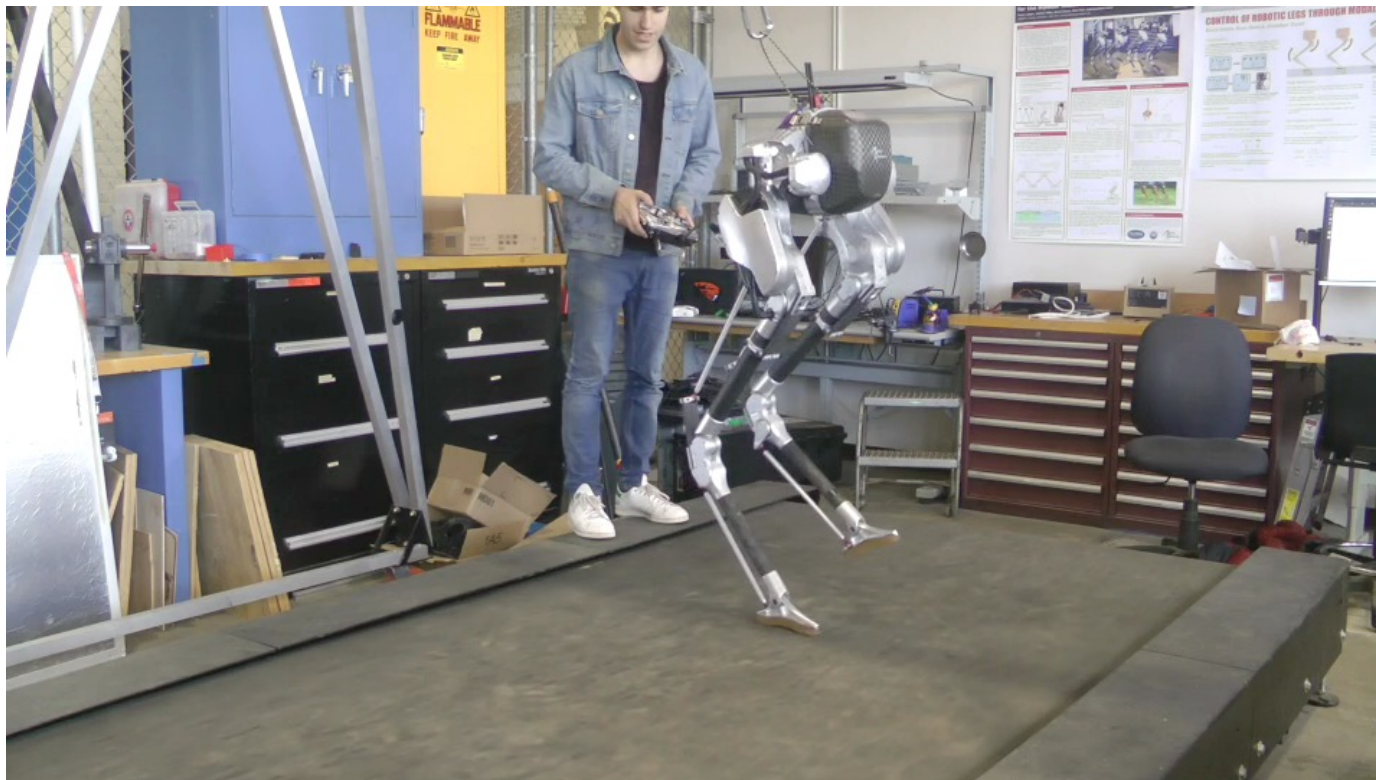
WALK AND STUMBLE



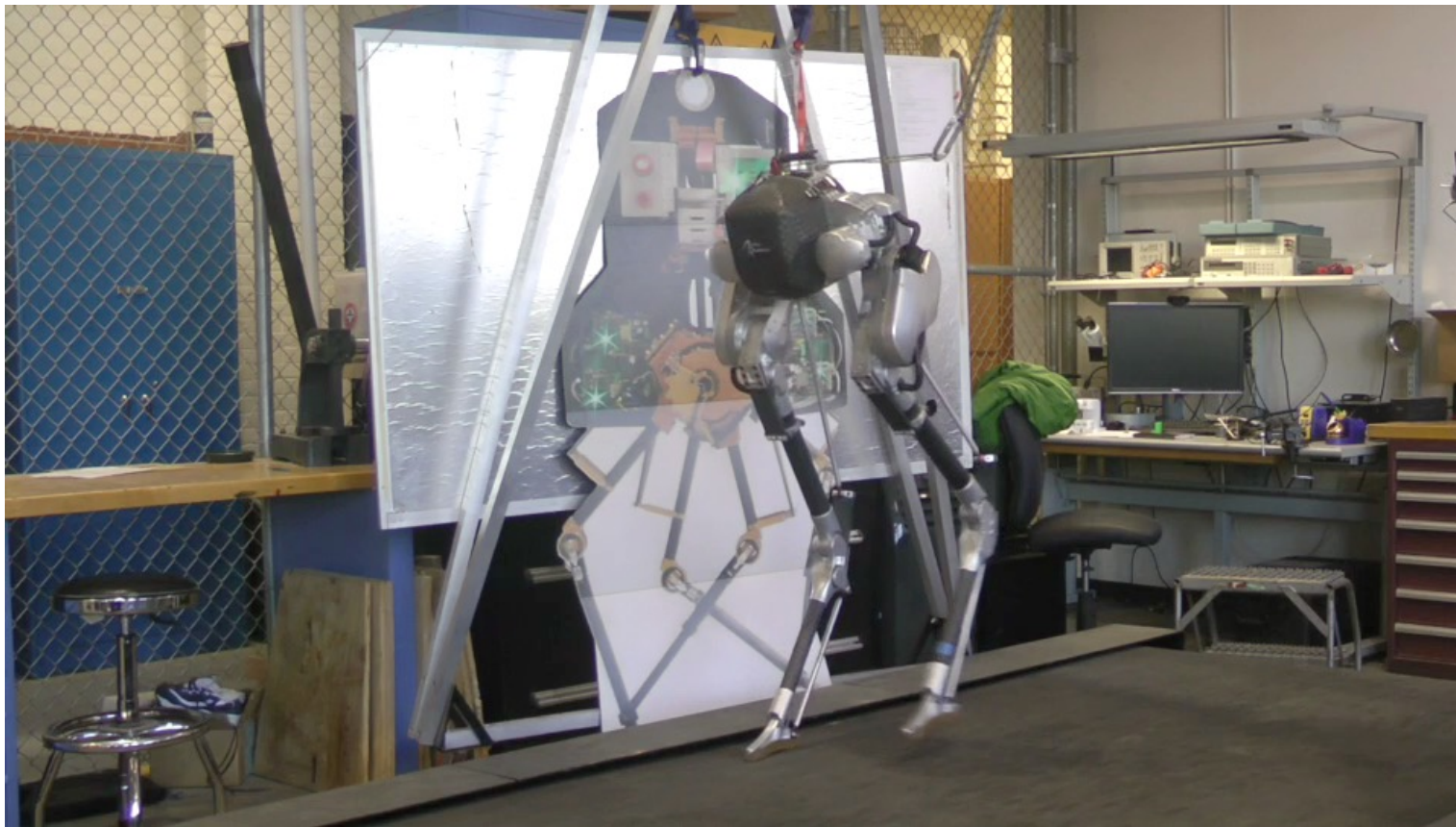
BACKWARDS WALK



SIDE-STEPS



SMOOTH STYLE



HIGH-STEPPING



PERTURBATION TEST

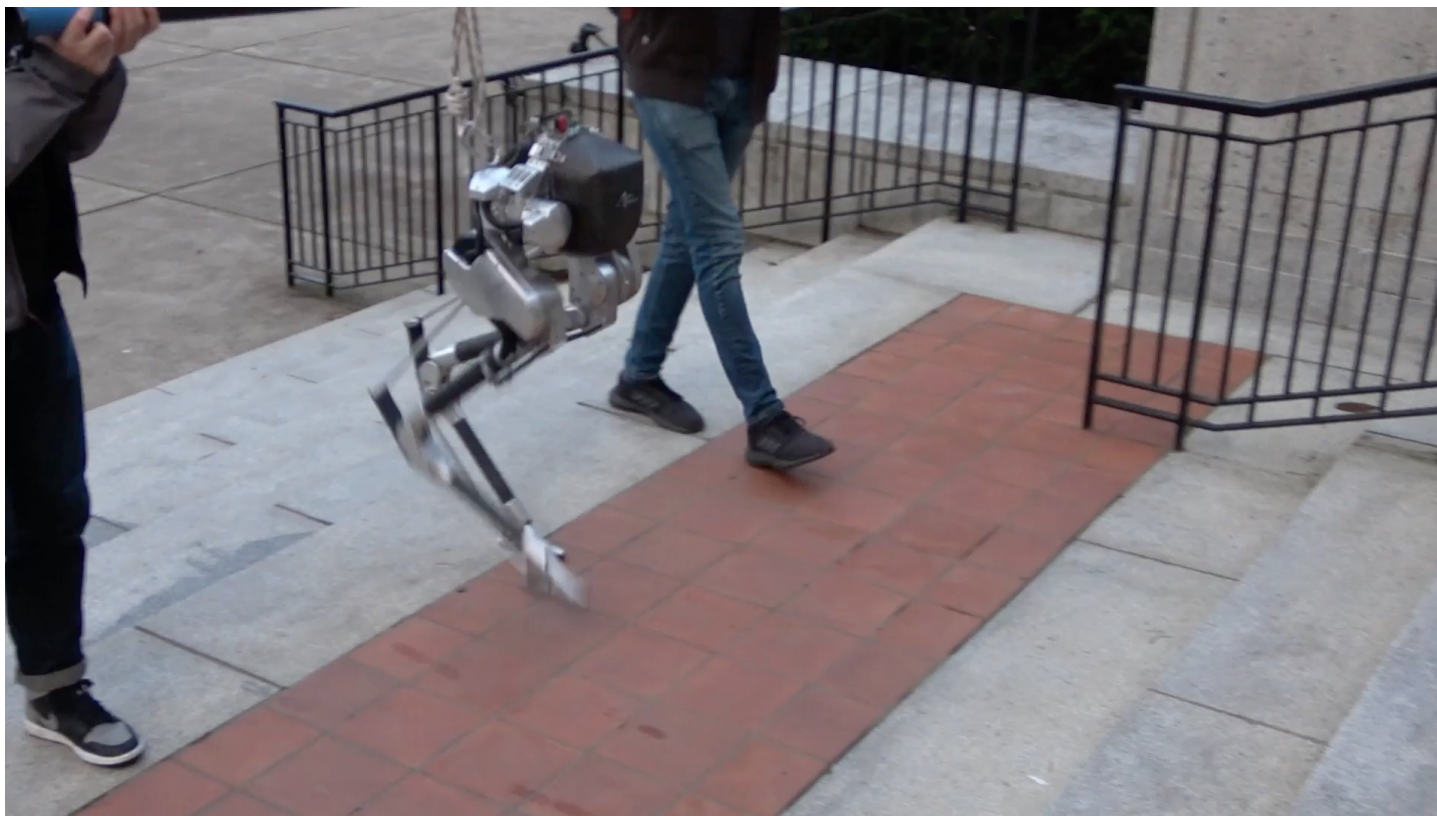
Perturbation Test

RL FOR CASSIE -- INSIGHTS

- objective: imitation, goal vel, body shake, foot height/speed
- action space: residual PD target angles
- sim-to-real
 - robustness from stochastic policy training
 - well-calibrated dynamics model
 - use the state estimator during training
 - no dynamics randomization required

SIM-TO-REAL APPROACHES

- dynamics randomization: robust “average” controller
- online learned system ID: efficient adaptation
 - explicit, e.g., Georgia Tech: W Yu, J Tan, CK Liu, G Turk (2017)
 - implicit, e.g., with memory
[Peng et al 2018] [OpenAI 2018, 2019]
- develop a good simulation model



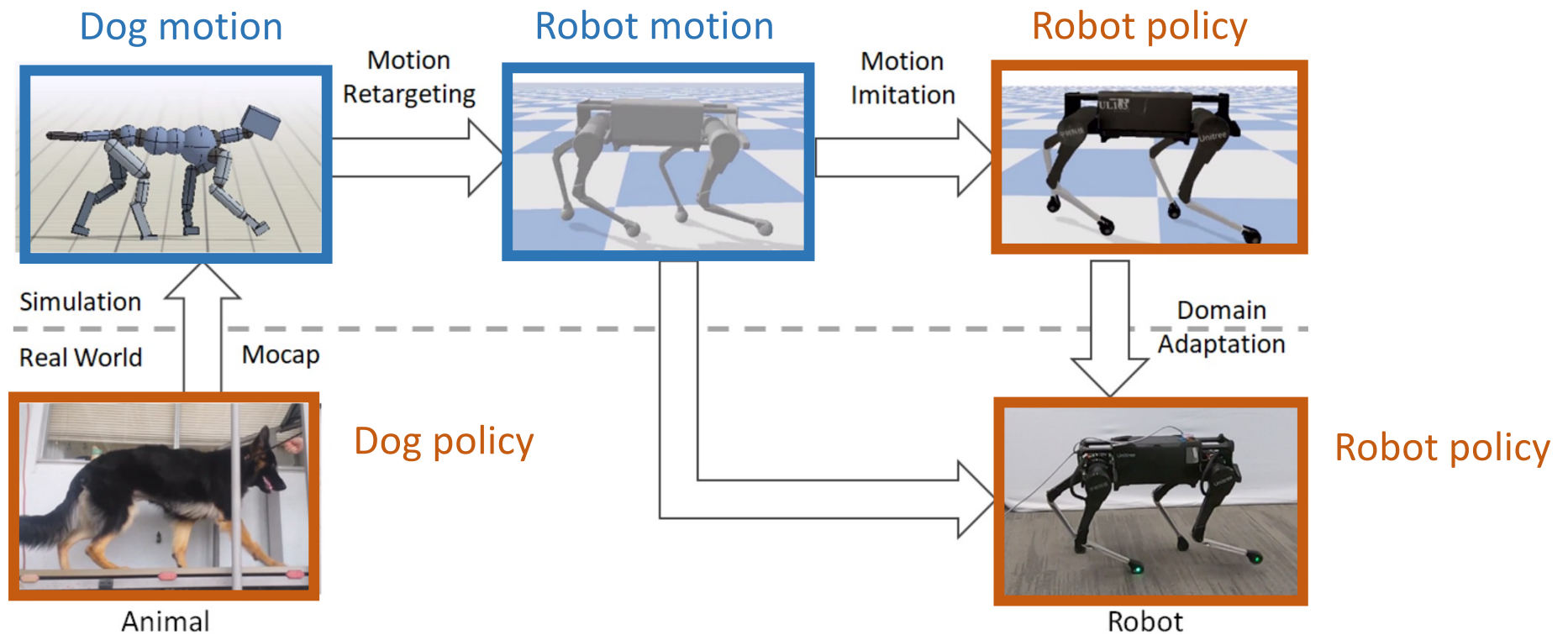
Blind Bipedal Stair Traversal via Sim-to-Real Reinforcement Learning

Jonah Siekmann^{*†}, Kevin Green^{*}, John Warila^{*}, Alan Fern^{*}, Jonathan Hurst^{*†}

^{*}CoRIS Institute, Oregon State University

RSS 2021

REAL - SIM - REAL



[“Learning Agile Robotic Locomotion Skills by Imitating Animals”, RSS 2020]

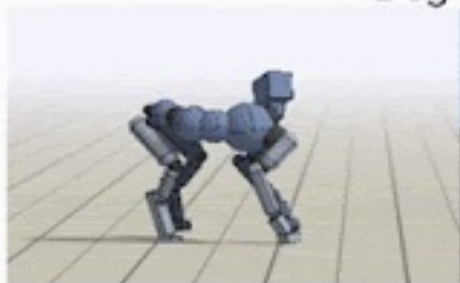
Dog Pace



Dog Trot



Dog Spin



Dog Backwards Trot



SKILLS FROM VIDEO: REINFORCEMENT LEARNING OF PHYSICAL SKILLS FROM VIDEO

[SIGGRAPH ASIA 2018]

Transactions on Graphics (Proc. ACM SIGGRAPH Asia 2018)

Xue Bin Peng

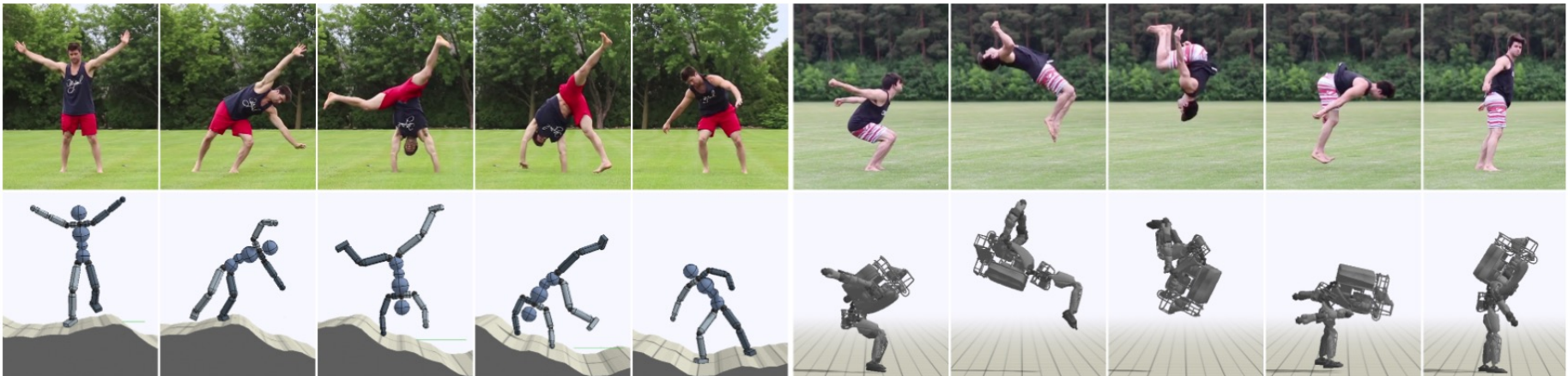
Angjoo Kanazawa

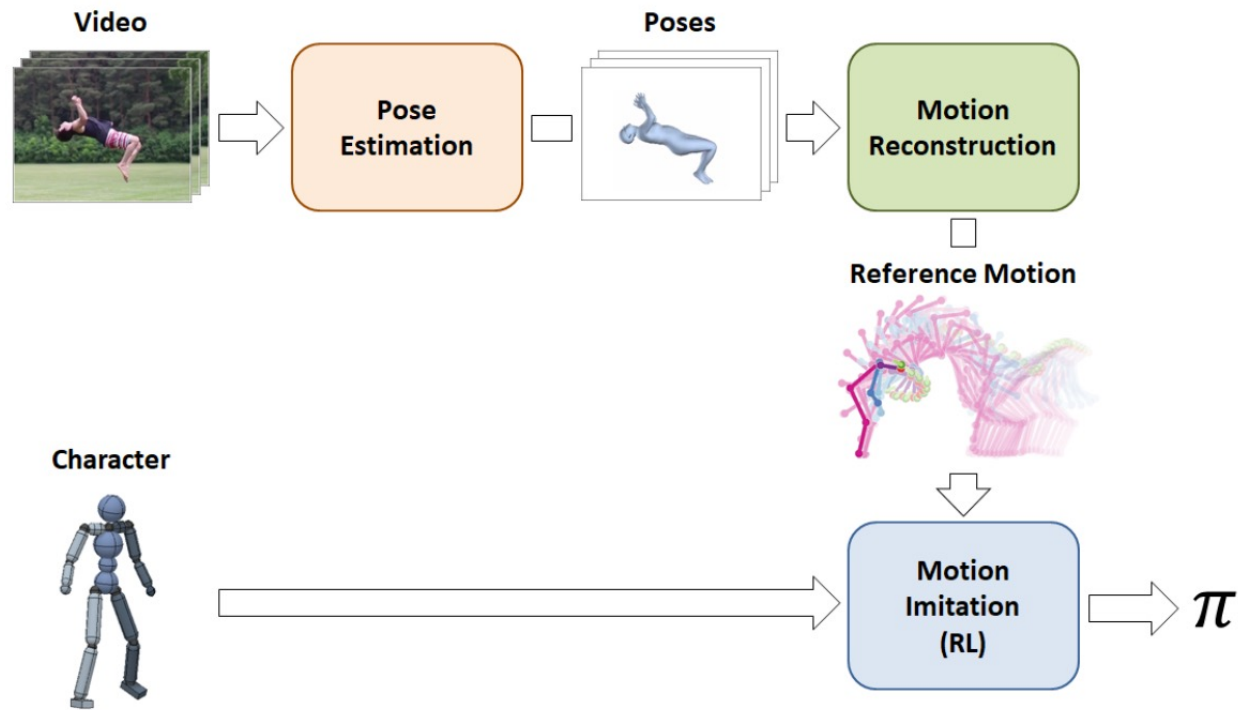
Jitendra Malik

Pieter Abbeel

Sergey Levine

University of California, Berkeley

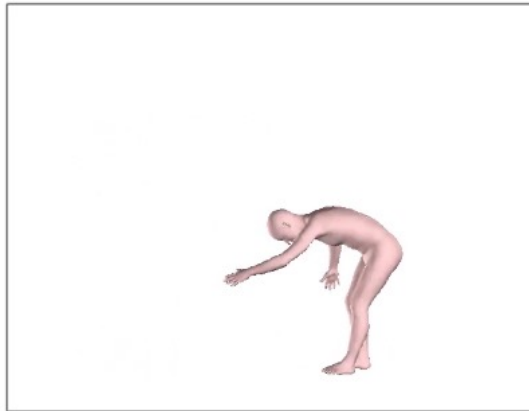




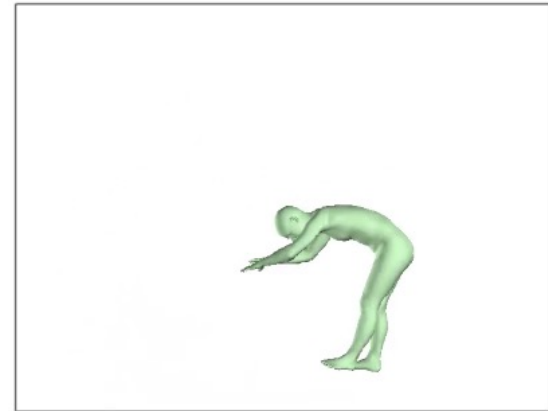
Pose Estimation



Video: Handspring A



No Augmentation
[Kanazawa et al. 2018]

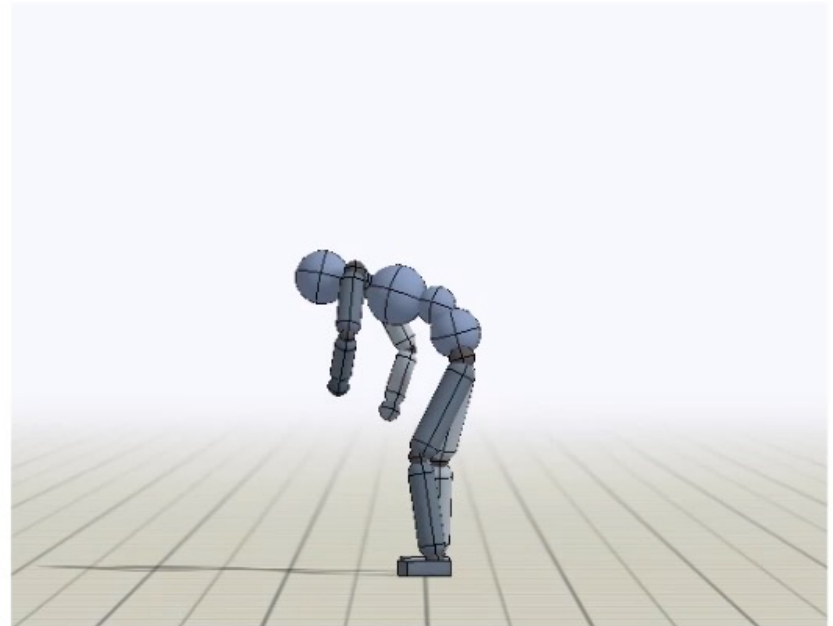


With Augmentation
(our work)

SKILLS FROM VIDEOS



Video



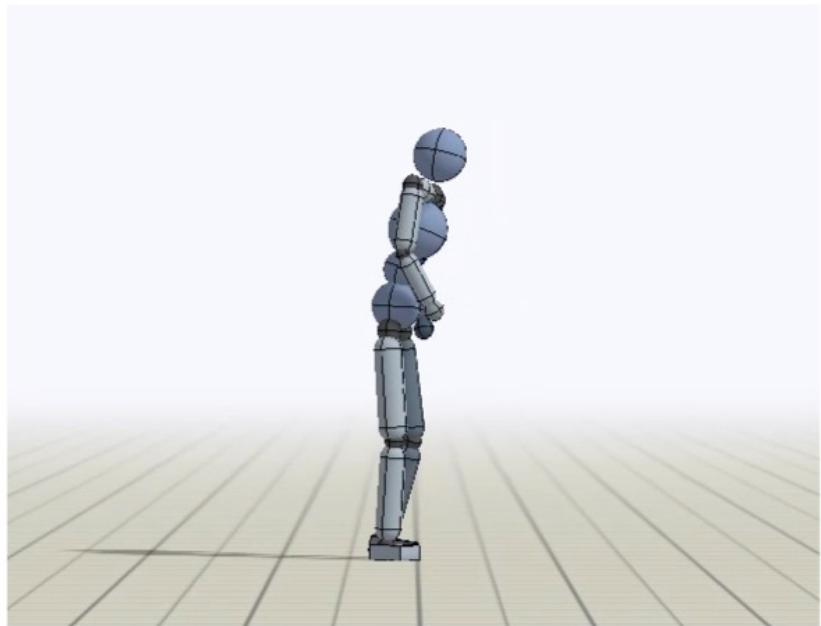
Simulation

["Skills from Video", SIGGRAPH ASIA 2018]

SKILLS FROM VIDEOS

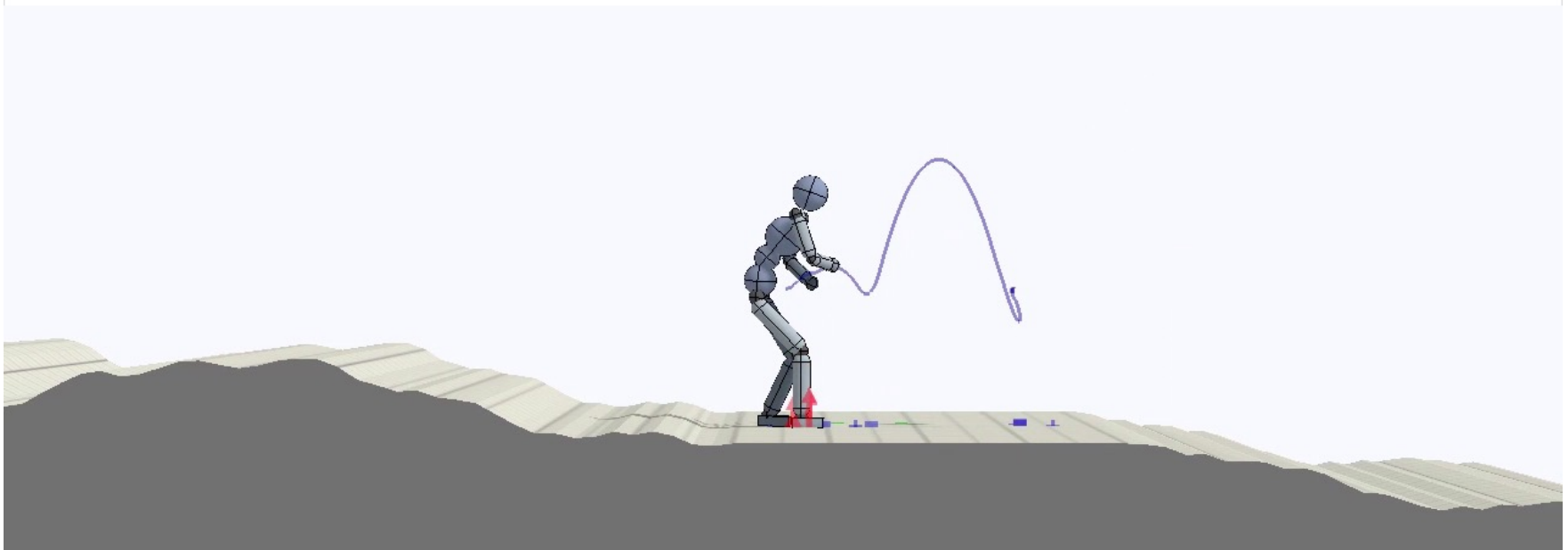


Video



Simulation

Retargeting

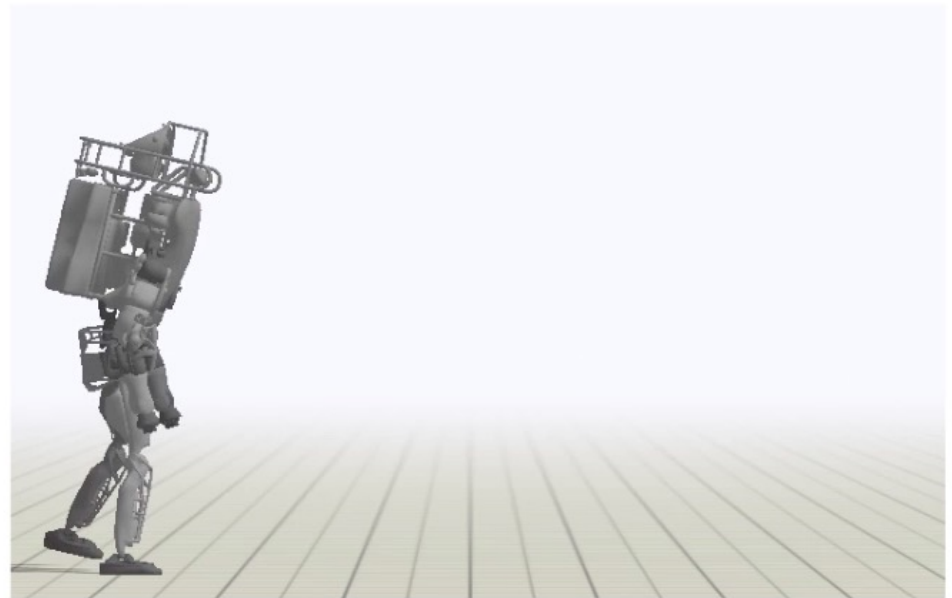


and retarget to different environments.

Character Retargeting



Video: Jump



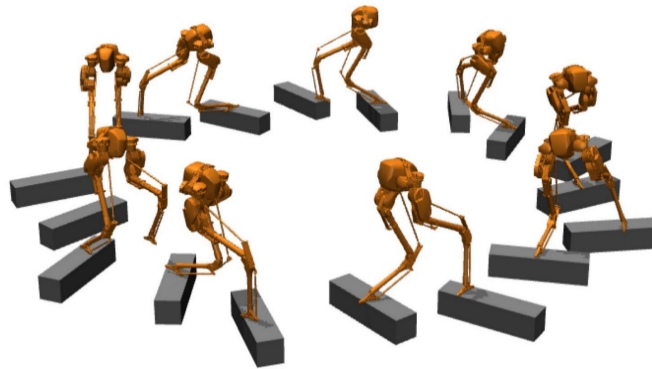
Policy

LESSONS LEARNED

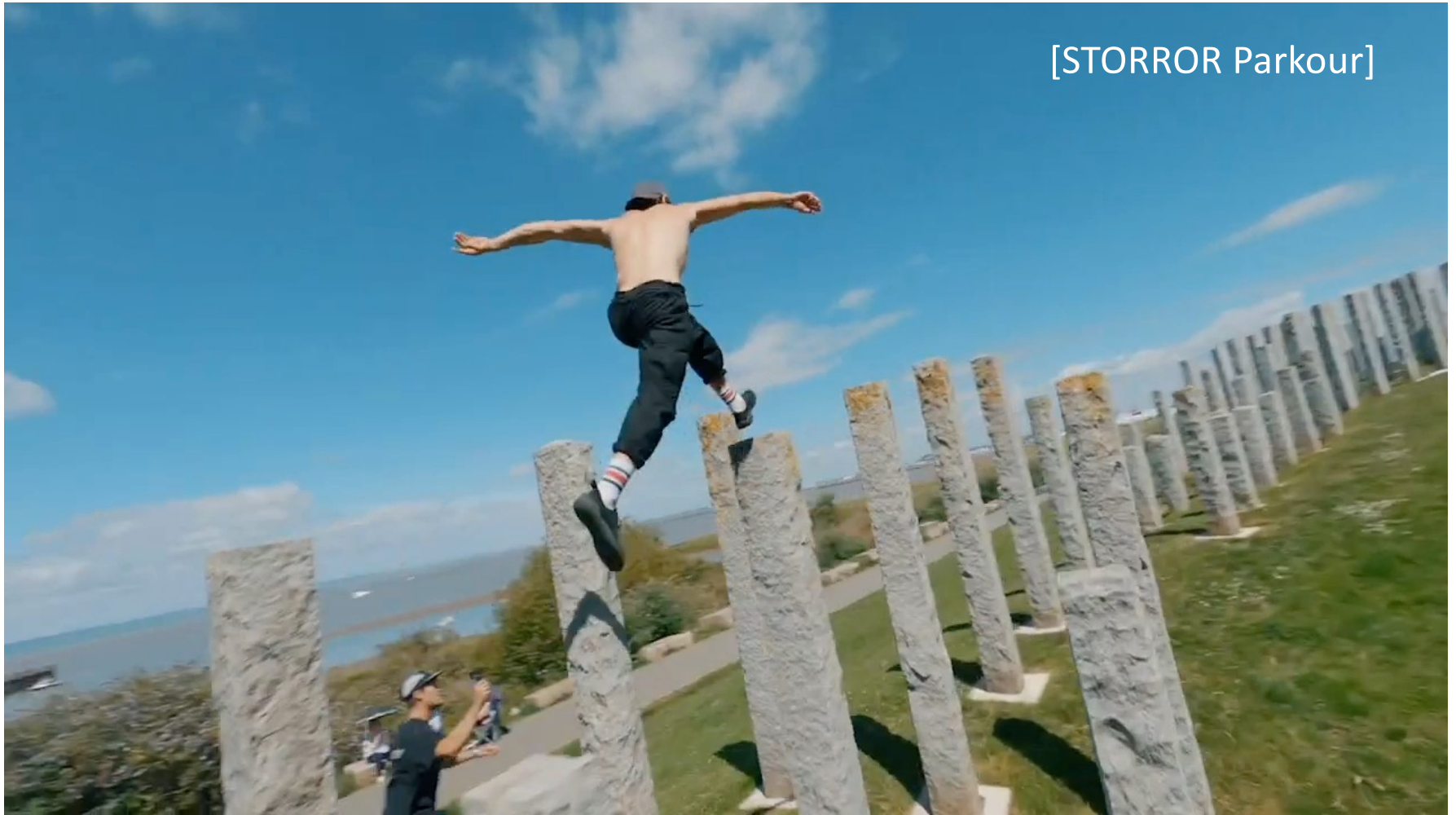
- physics-based imitation is a good regularizer
- vision (analysis) - control (synthesis): mutual bootstrapping

STEPPING STONE LOCOMOTION USING RL

ALLSTEPS: Curriculum driven learning of stepping stone skills, SCA 2020
Zhaoming Xie, Hung Yu Ling, Nam Hee Kim, Michiel van de Panne



[STORROR Parkour]



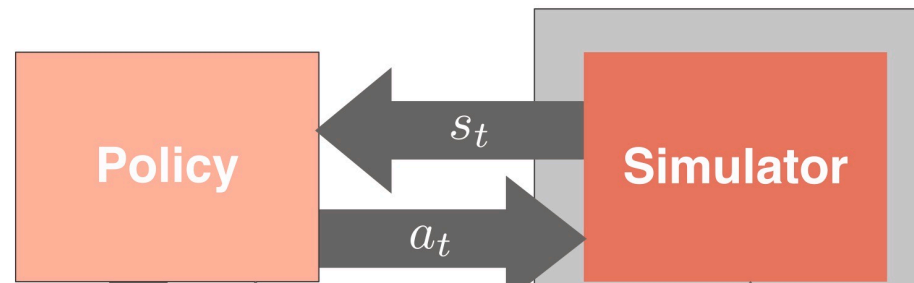
REWARD FUNCTION



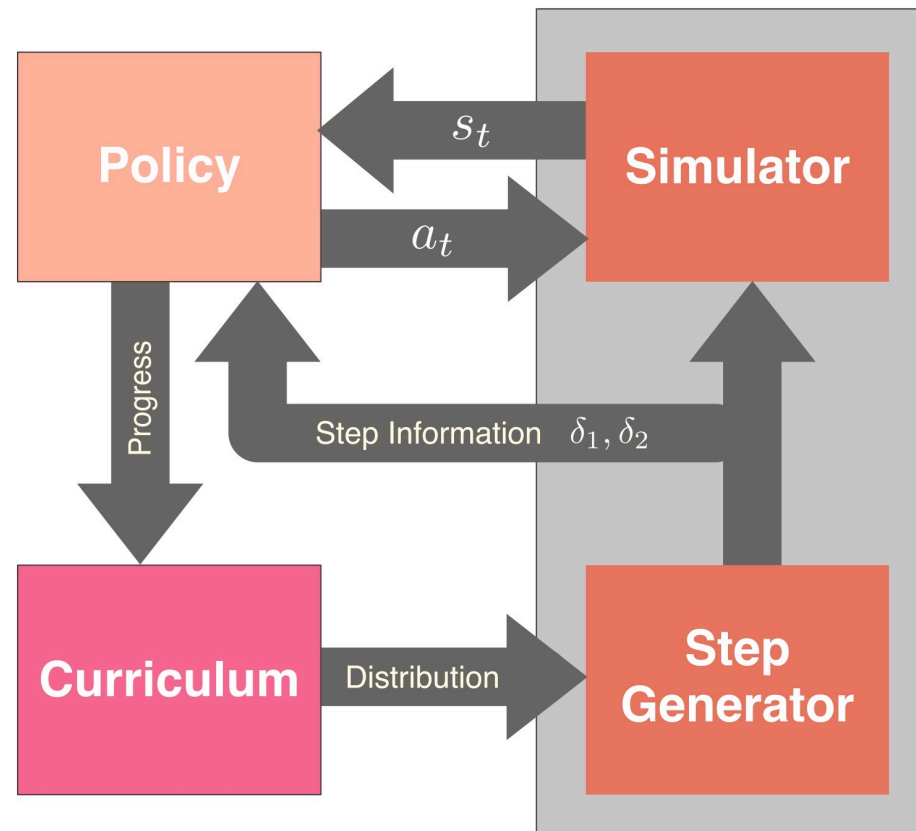
- forward progress & energy
- make contact with target step
- upright posture
- symmetry
- avoid joint limits

21 internal torques
two-step look-ahead

CURRICULUM-DRIVEN LEARNING

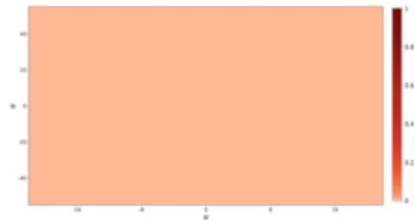


CURRICULUM-DRIVEN LEARNING

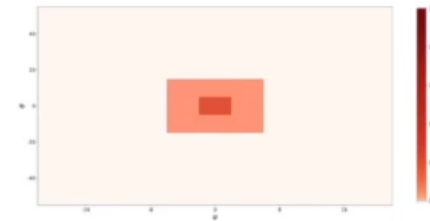


CURRICULUM-DRIVEN LEARNING

(1) Uniform Sampling
(baseline; curriculum-free)



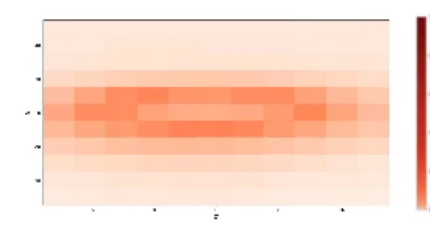
(2) Fixed Order



(3) Fixed Order Boundary



(4) Adaptive

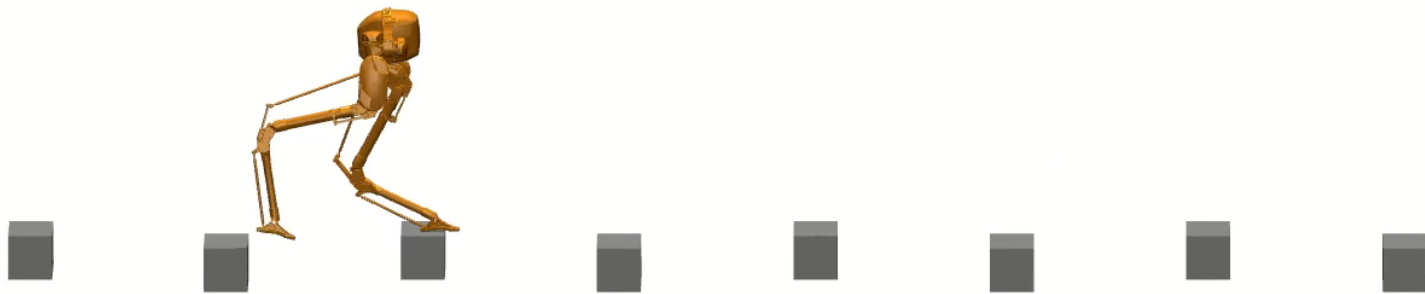


Practice steps that are neither too easy nor too hard!
~6 hours to compute on 12 cores

Long Steps

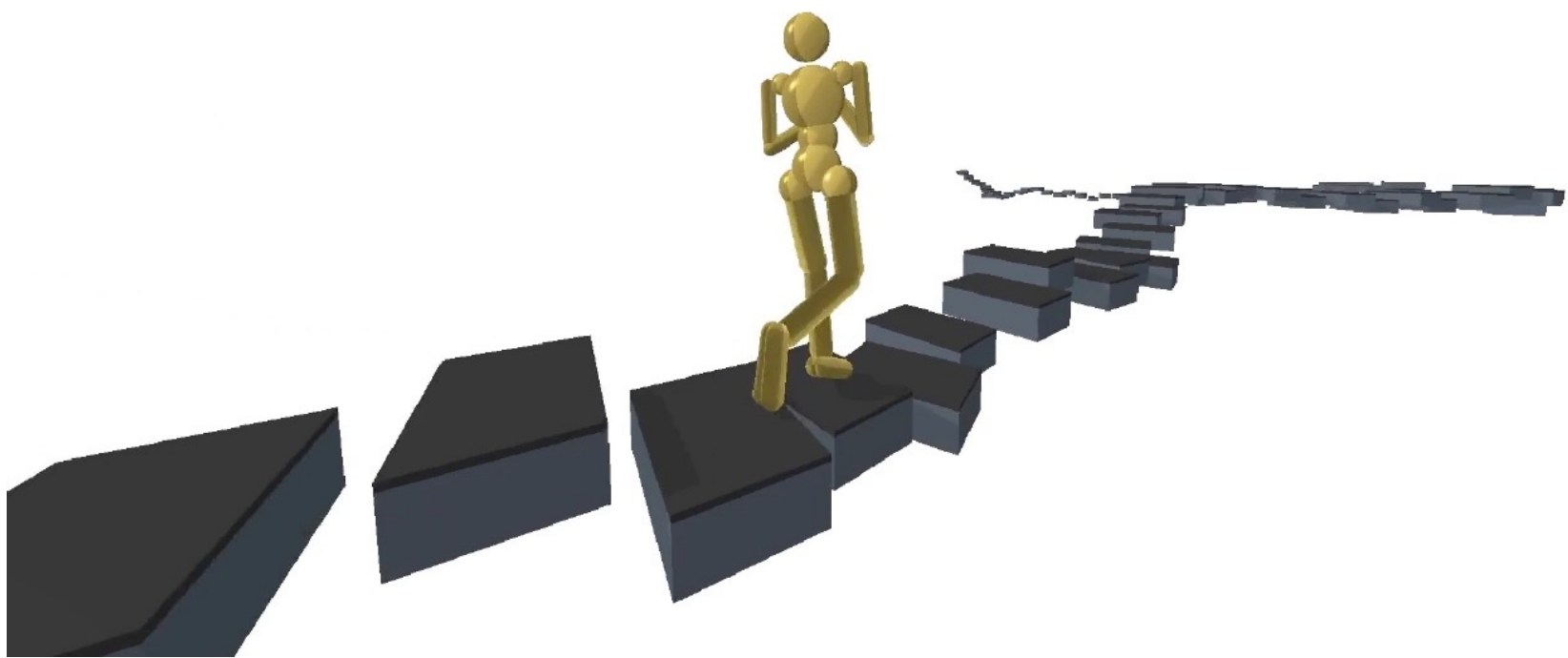


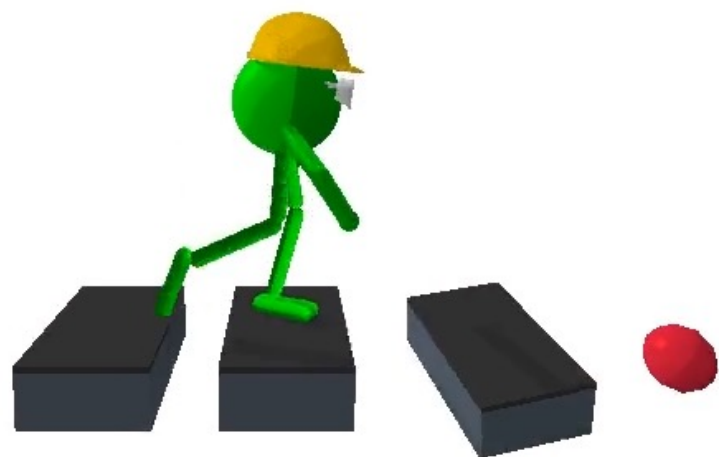
Long Steps



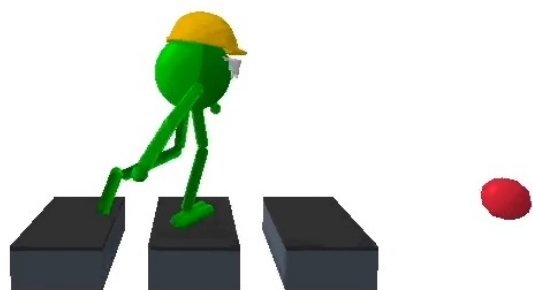
The simulation sees terrain, while the policy sees steps











LIMITATIONS

- missing: jumps, side-steps, sharp turns
- apriori foot-step plan
- quadrupeds?

RL FOR DISCOVERY OF JUMPING STRATEGIES

Discovering Diverse Athletic Jumping Strategies

[arXiv, SIGGRAPH 2021]

ZHIQI YIN, Simon Fraser University, Canada

ZESHI YANG, Simon Fraser University, Canada

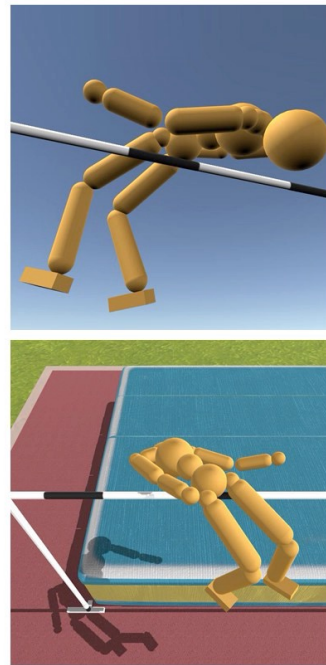
MICHIEL VAN DE PANNE, University of British Columbia, Canada

KANGKANG YIN, Simon Fraser University, Canada



Fosbury Flop

(2.0 m)

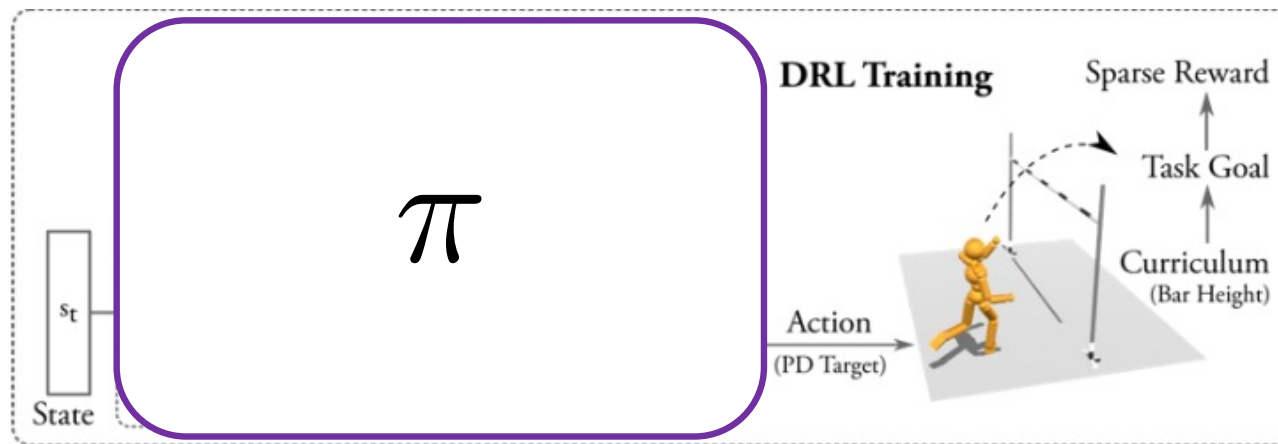


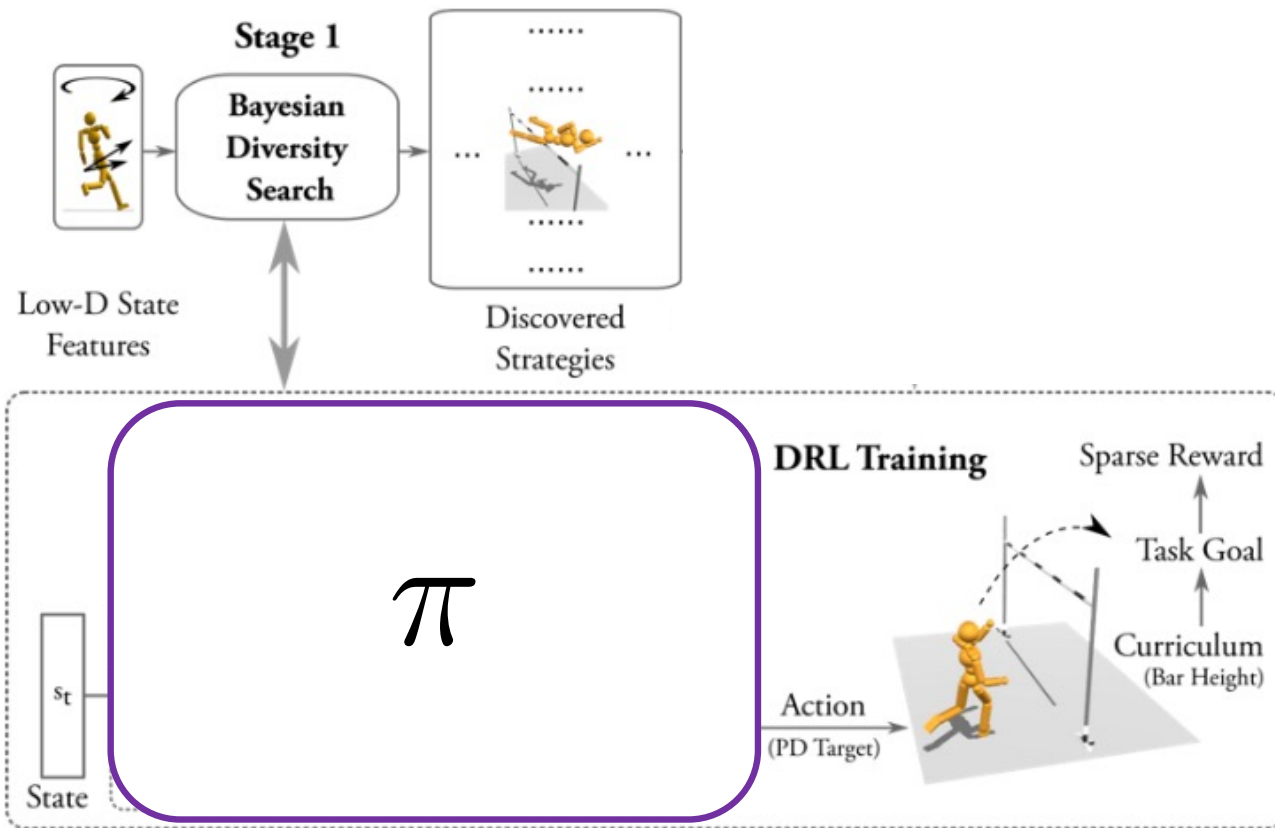
Stochastic Evaluation of High Jump Controller

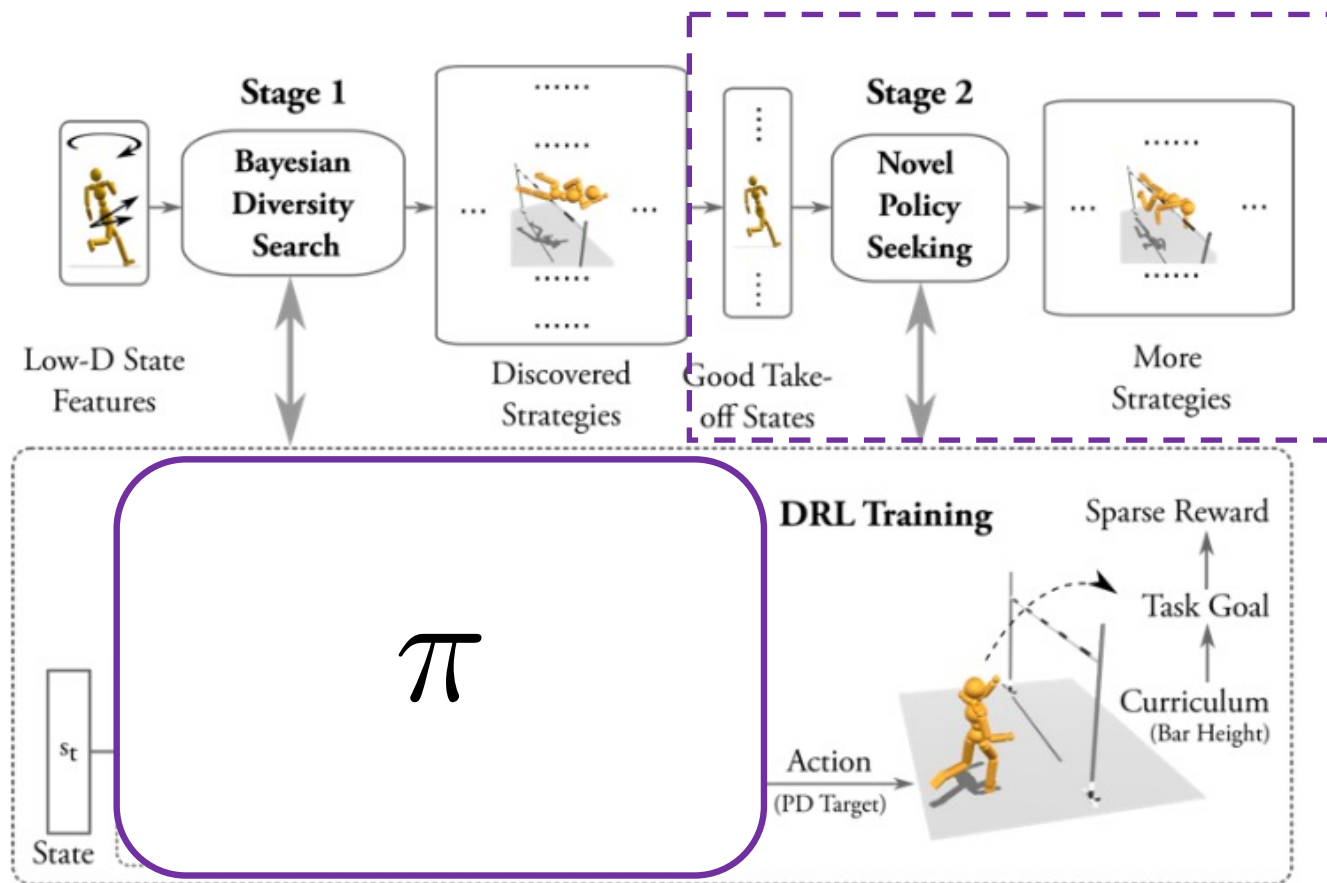
LEARNING CURRICULUM

Given: Take-off state

Learn: state-indexed control policy









(a) Straddle – max height=190cm



(b) Front Kick – max height=180cm



(c) Western Roll (facing up) – max height=160cm



(d) Scissor Kick – max height=150cm

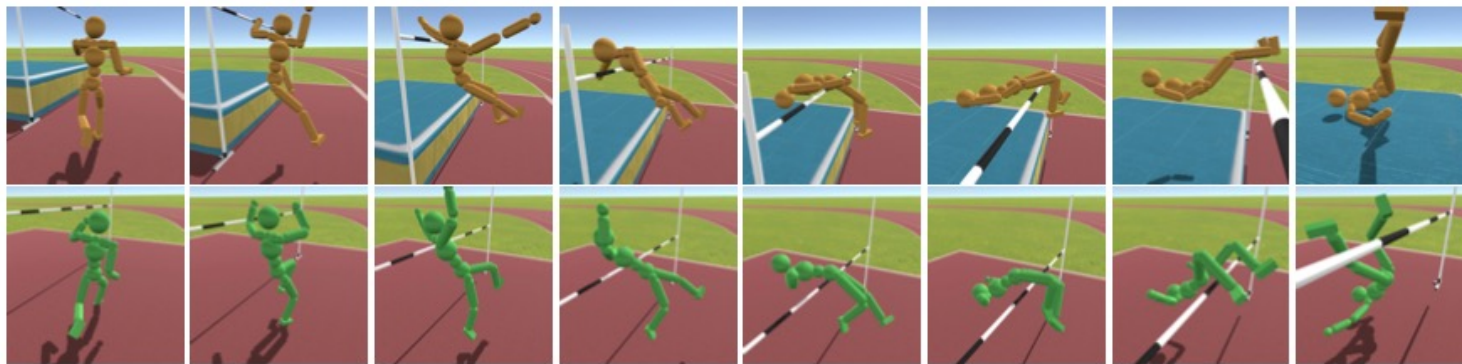


(e) Side Dive – max height=130cm

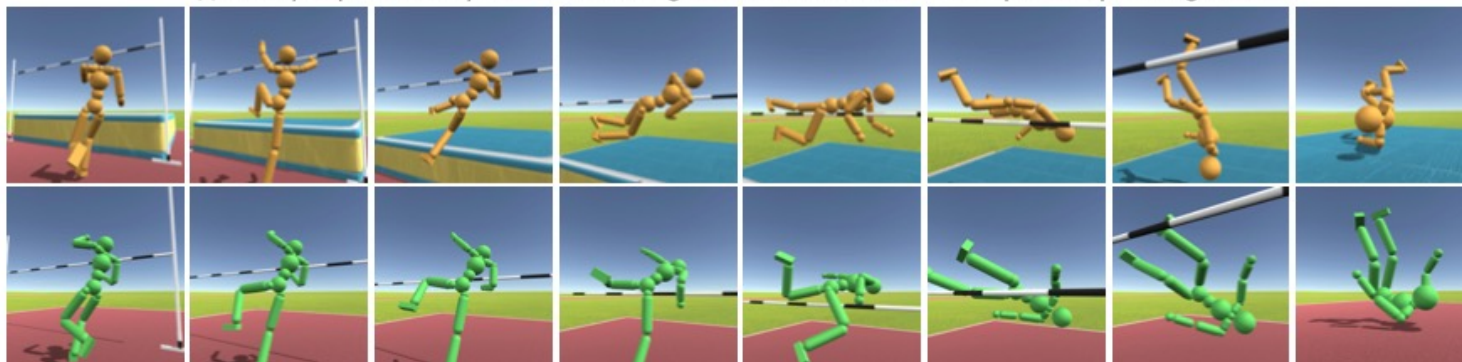


(f) Side Jump – max height=110cm

COMPARISON WITH MOCAP



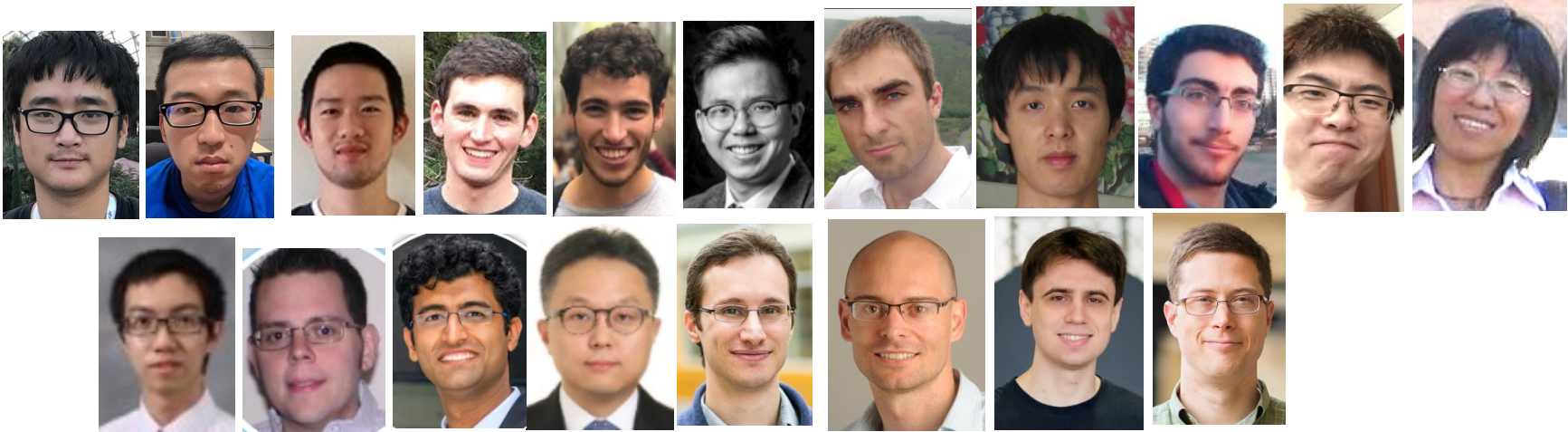
(a) Fosbury Flop. First row: synthesized – max height=200cm; Second row: motion capture – capture height=130cm.



(b) Straddle. First row: synthesized – max height=195cm; Second row: motion capture – capture height=130cm.

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QUESTIONS & DISCUSSION

